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- (54) **CONTROL OF POWER USE OF A DEVICE POWERED BY A COMMUNICATION BUS AND DETERMINING SIDEBAND SIGNALING VOLTAGE LEVEL**
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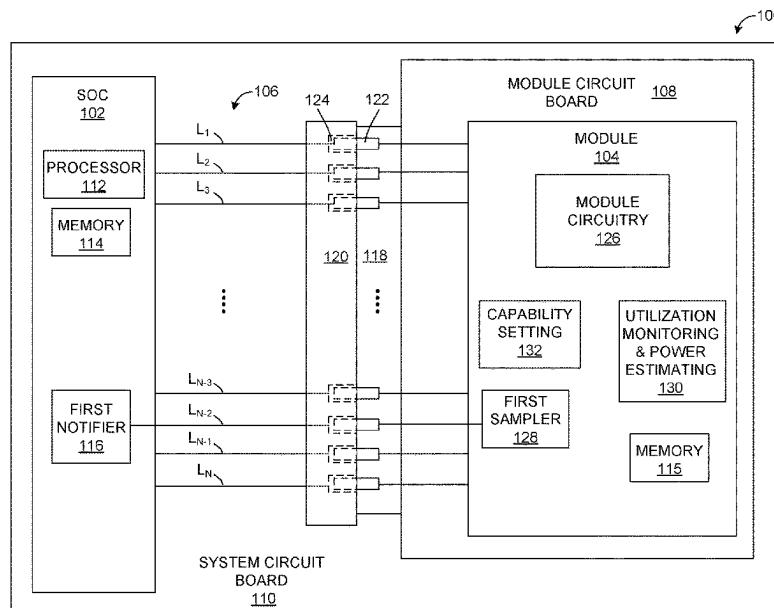
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(57) **ABSTRACT**

An apparatus comprises a circuit board comprising a connector in or on the circuit board. The apparatus is to be coupled to a bus via the connector. The apparatus comprises first circuitry to communicate with a processor via the connector and bus, second circuitry to detect a utilization state of the first circuitry, determine, based on the detected utilization state of the first circuitry, a level of current to be conducted with the connector; and generate a signal indicating of whether the level of current exceeds a threshold current capacity of the connector, and third circuitry to select a first operational mode from among multiple operational modes. The first circuitry is to operate in any of the multiple operational modes responsive to the third circuitry. Additional circuitry may be provided to identify a voltage level capability of the apparatus and to configure voltage level shifting circuitry connected to the bus.

20 Claims, 8 Drawing Sheets



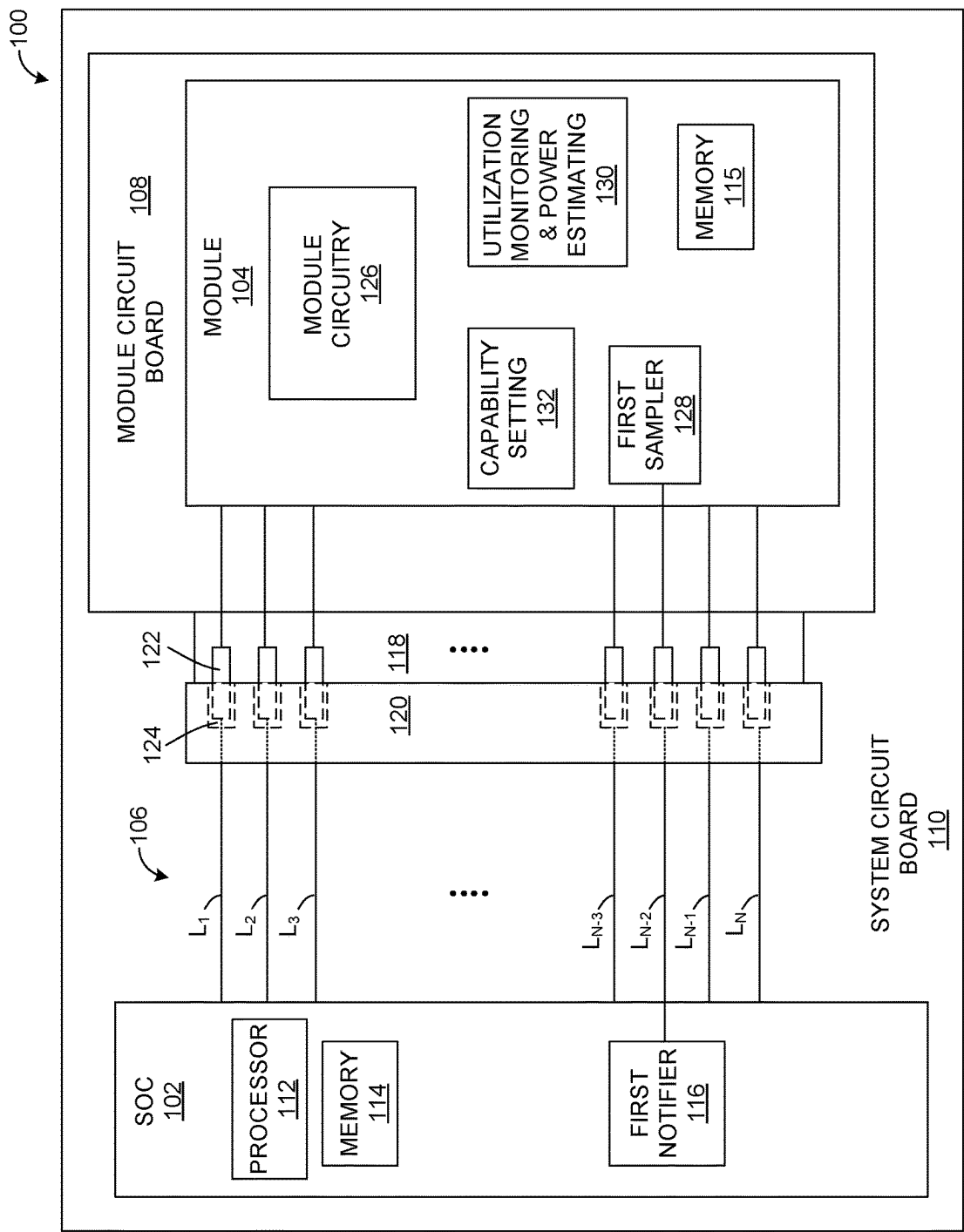


Fig. 1

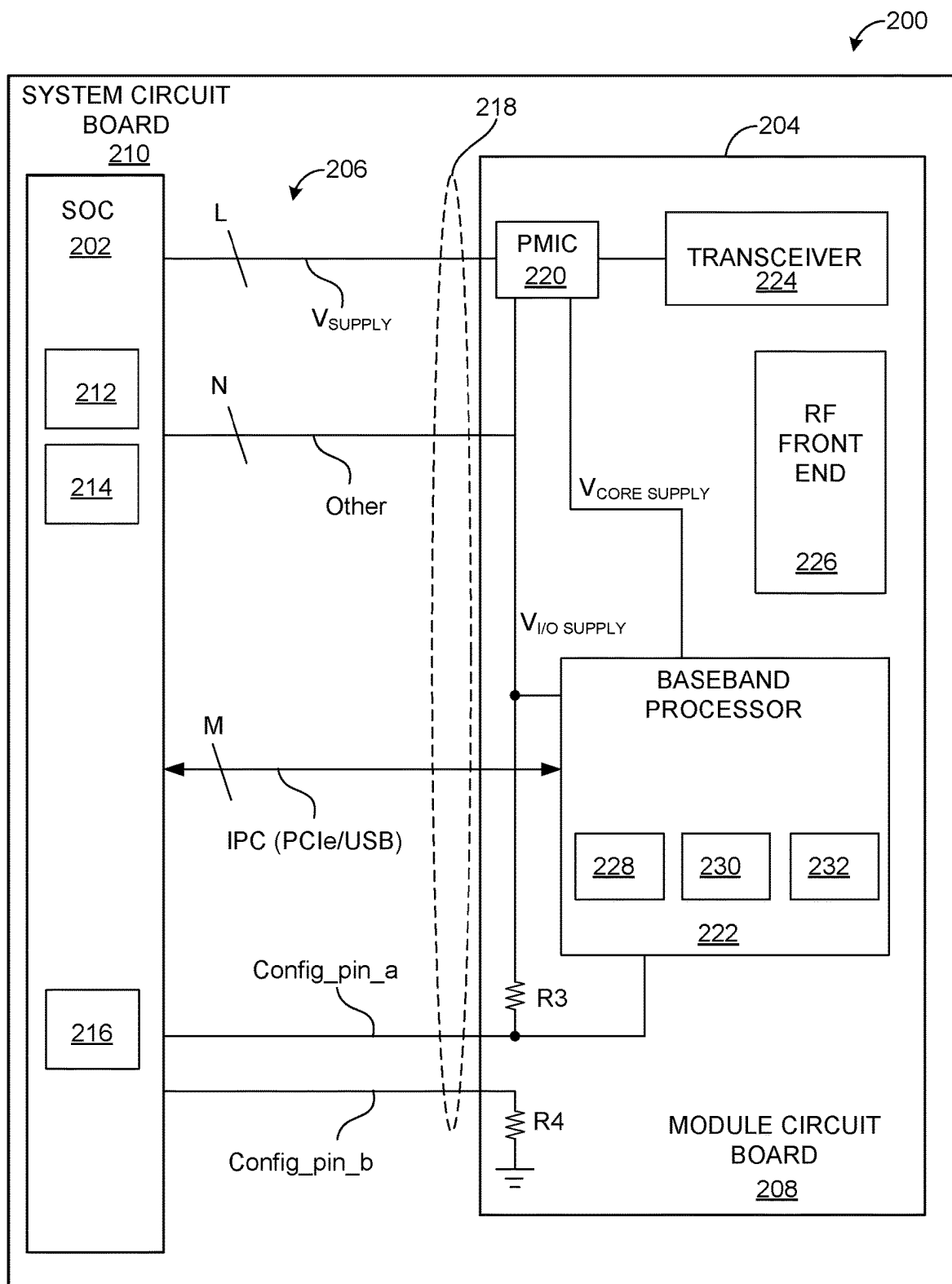


Fig. 2

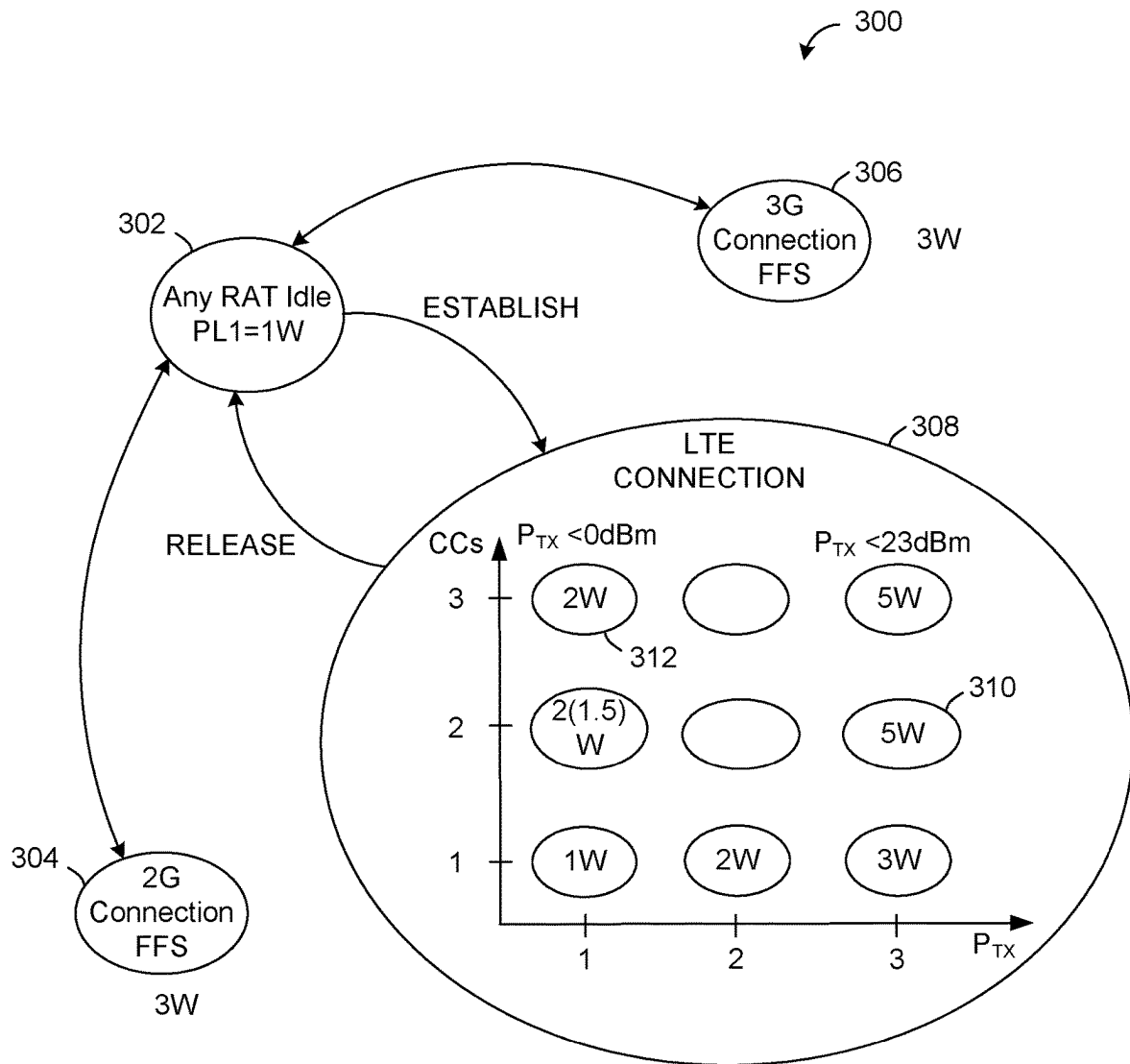
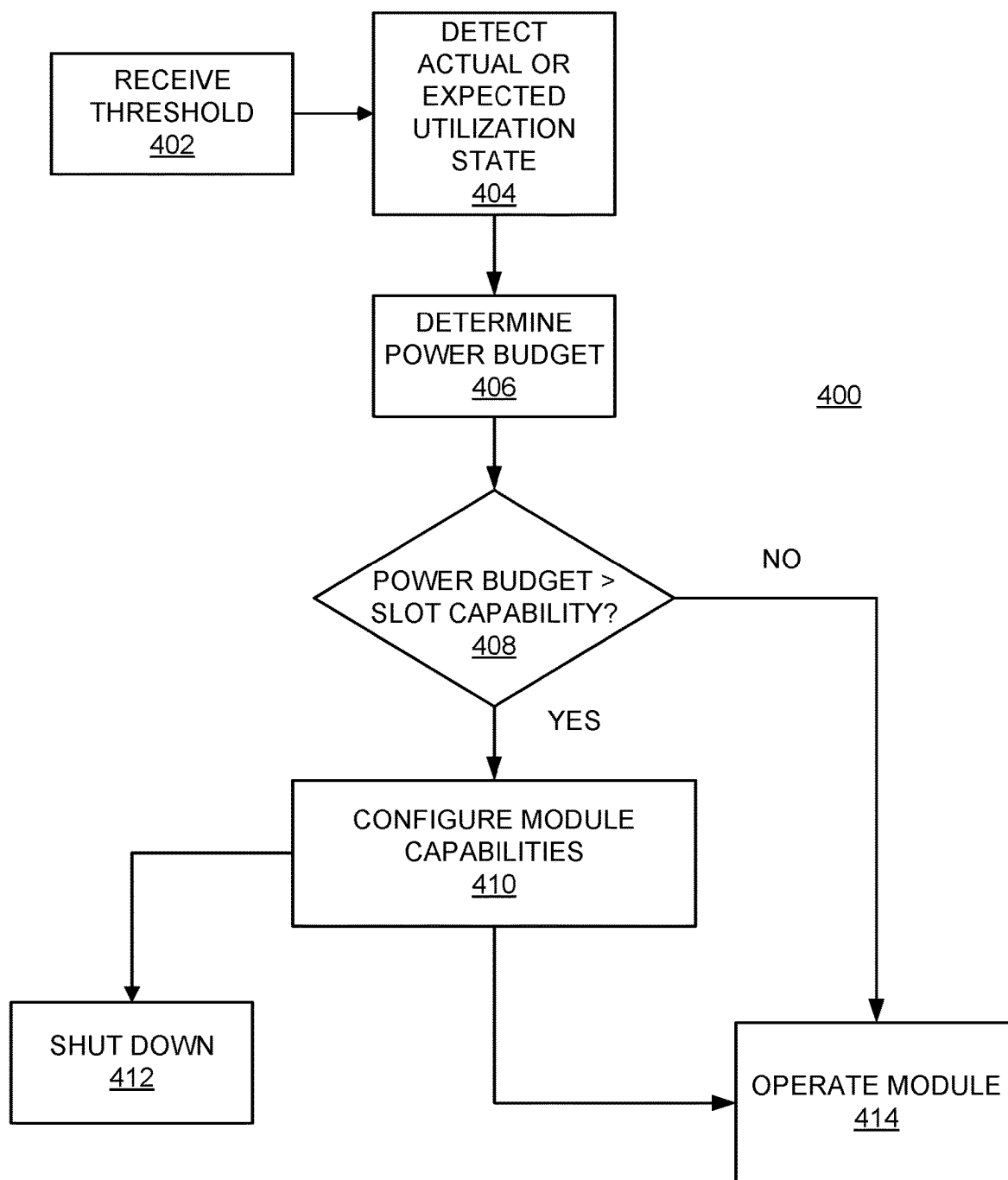


Fig. 3

**Fig. 4**

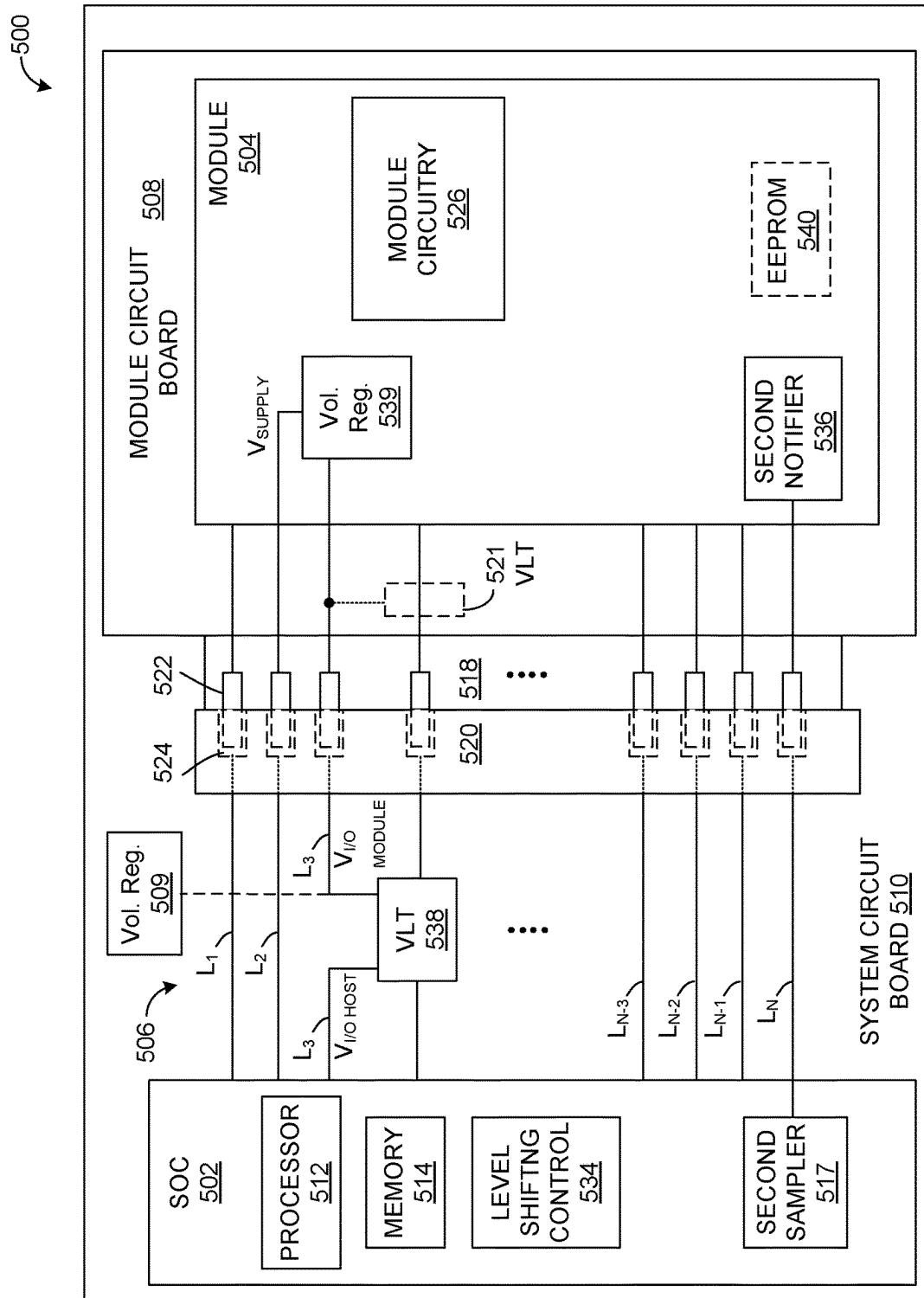
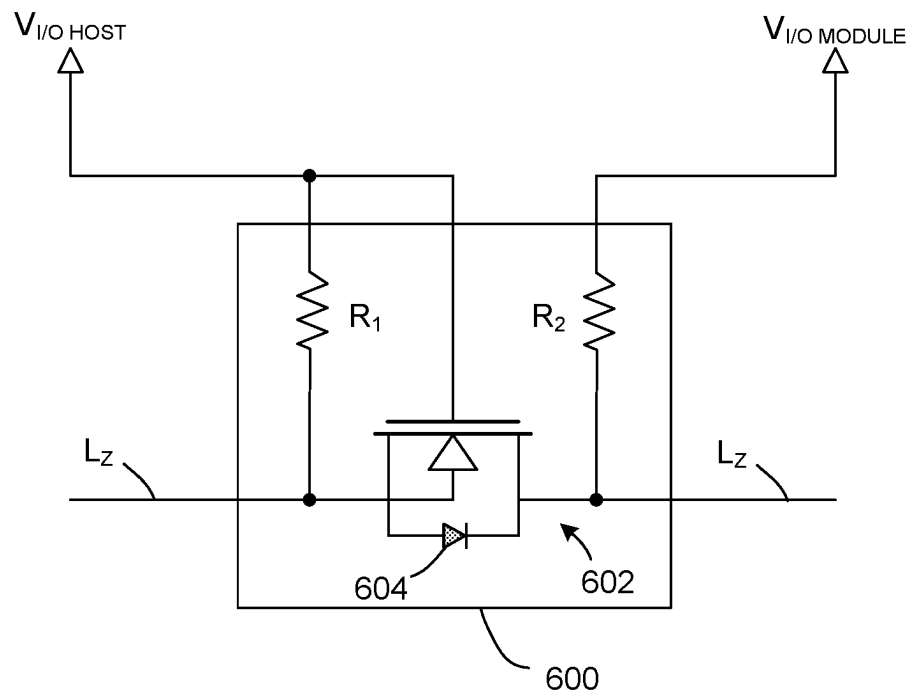


Fig. 5

**Fig. 6**

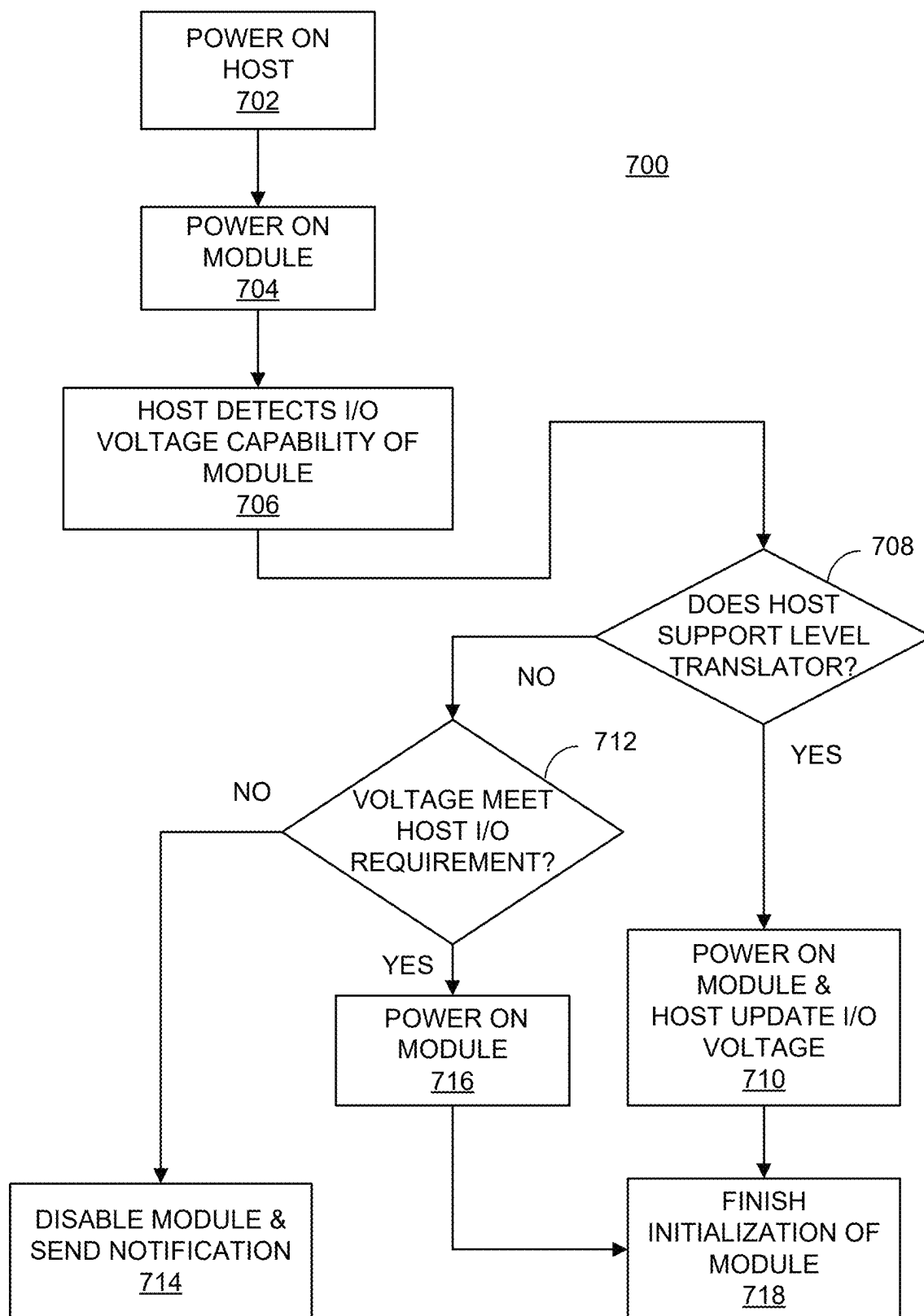


Fig. 7

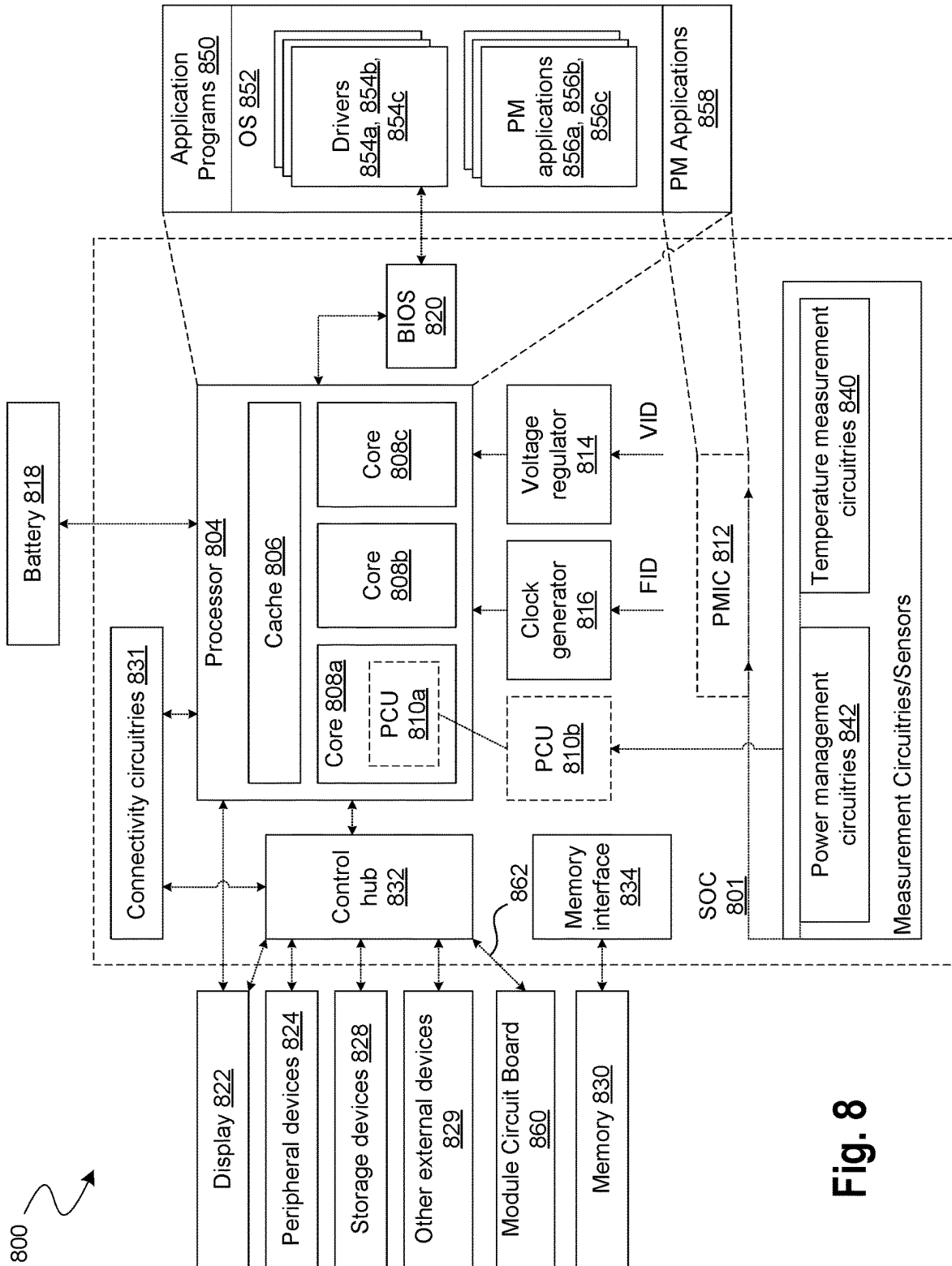


Fig. 8

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CONTROL OF POWER USE OF A DEVICE POWERED BY A COMMUNICATION BUS AND DETERMINING SIDEBAND SIGNALING VOLTAGE LEVEL

BACKGROUND

Modern computer systems come in many forms, such as servers, desktop units, laptops, and smartphones. A computer system may include many components, such as a central processing unit (CPU), memory, input/output (I/O) devices, such as a keyboard and display, specialized chips for performing operations like graphics processing or communications, and various other components. Computer systems may include a system-on-a-chip (SOC) that integrates many of the components onto a single integrated circuit (IC) or a system-in-package (SIP) that integrates two or more ICs in a single package. Some computer systems include an add-in module or expansion card. In all forms, computer systems include one or more busses to transfer data between devices or components within the computer system.

BRIEF DESCRIPTION OF THE DRAWINGS

The embodiments of the disclosure will be understood more fully from the detailed description given below and from the accompanying drawings of various embodiments of the disclosure, which, however, should not be taken to limit the disclosure to the specific embodiments, but are for explanation and understanding only.

FIG. 1 is a functional block diagram illustrating elements of a computing system to limit electric current consumption of a module that receives power from a bus in accordance with some embodiments.

FIG. 2 is a functional block diagram illustrating elements of a computing system to limit electric current consumption of a wireless wide area network (WWAN) module that receives power from a bus in accordance with some embodiments.

FIG. 3 illustrates a state diagram for limiting electric current consumption of a WWAN module in accordance with some embodiments.

FIG. 4 illustrates a flow diagram of a process for limiting current in a module according to various embodiments.

FIG. 5 is a functional block diagram illustrating elements of a computing system to assure that the voltage level used by a module and by a host for sideband signaling match in accordance with some embodiments.

FIG. 6 is a block diagram of voltage level shifter circuitry in accordance with some embodiments.

FIG. 7 illustrates a flow diagram of a process for assuring that the voltage level used by a module for sideband signaling will be compatible with voltage level used by a host according to various embodiments.

FIG. 8 illustrates a computer system or computing device to limit electric current consumption of a module that receives power from a bus, and to assure that the voltage level used by the module and by a host for sideband signaling match, in accordance with some embodiments.

DETAILED DESCRIPTION

In a computer system, a module, such as an add-in module or expansion card, may be connected to bus via module connector and socket or slot on the mother board of the system. An add-in module adds functionality to the computer system. It is important that the module be operated in

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a way that is compatible with various specifications of the computer system, such as a specification for the bus. An add-in module may be connected at the time of manufacturing by a board manufacturer, a module manufacturer, or by an original equipment manufacturer (OEM). In addition, a module may be connected by an end user subsequent to manufacture. Because the party that connects the module to the bus may not be aware of whether the generation of module is compatible with a bus or other specification, it is possible that the party will insert a non-compliant generation of add-on card into socket of a system circuit board. If the specification relates to a threshold current capacity per pin, excessive heat and reliability issues may result. If the specification relates to voltage ranges used for sideband signals, a mismatch in the sideband signaling voltage may lead to the electrical stress in the host or add-card.

Various embodiments relate generally to computing systems that limit electric current consumption of a module that receives power from a bus when the module operates in a mode or is expected to operate in a mode in which electric current consumption may exceed a threshold. Accordingly, an advantage of some embodiments is that the computer system does not generate excessive heat and or suffer from reliability issues.

Other embodiments relate generally to computing systems that include a mechanism to assure that a voltage level used by a module, e.g., an add-in module, and by a host for sideband signaling match. Accordingly, an advantage of some embodiments is that the computer system sideband signals of the module and host are compatible, and electrical stress in the host or module are avoided.

In the following description, numerous details are discussed to provide a more thorough explanation of the embodiments of the present disclosure. It will be apparent to one skilled in the art, however, that embodiments of the present disclosure may be practiced without these specific details. In other instances, well-known structures and devices are shown in block diagram form, rather than in detail, in order to avoid obscuring embodiments of the present disclosure.

Note that in the corresponding drawings of the embodiments, signals are represented with lines. Some lines may be thicker, to indicate a greater number of constituent signal paths, and/or have arrows at one or more ends, to indicate a direction of information flow. Such indications are not intended to be limiting. Rather, the lines are used in connection with one or more exemplary embodiments to facilitate easier understanding of a circuit or a logical unit. Any represented signal, as dictated by design needs or preferences, may actually comprise one or more signals that may travel in either direction and may be implemented with any suitable type of signal scheme.

Throughout the specification, and in the claims, the term “connected” means a direct connection, such as electrical, mechanical, or magnetic connection between the things that are connected, without any intermediary devices. The term “coupled” means a direct or indirect connection, such as a direct electrical, mechanical, or magnetic connection between the things that are connected or an indirect connection, through one or more passive or active intermediary devices. The term “circuit” or “module” may refer to one or more passive and/or active components that are arranged to cooperate with one another to provide a desired function. A module may comprise a circuit or circuitry, as defined below. The term “signal” may refer to at least one current signal, voltage signal, magnetic signal, or data/clock signal. The

meaning of “a,” “an,” and “the” include plural references. The meaning of “in” includes “in” and “on.”

The term “device” may generally refer to an apparatus according to the context of the usage of that term. For example, a device may refer to a stack of layers or structures, a single structure or layer, a connection of various structures having active and/or passive elements, etc. Generally, a device is a three-dimensional structure with a plane along the x-y direction and a height along the z direction of an x-y-z Cartesian coordinate system. The plane of the device may also be the plane of an apparatus which comprises the device.

The term “scaling” generally refers to converting a design (schematic and layout) from one process technology to another process technology and subsequently being reduced in layout area. The term “scaling” generally also refers to downsizing layout and devices within the same technology node. The term “scaling” may also refer to adjusting (e.g., slowing down or speeding up—i.e. scaling down, or scaling up respectively) of a signal frequency relative to another parameter, for example, power supply level.

The terms “substantially,” “close,” “approximately,” “near,” and “about,” generally refer to being within $\pm 10\%$ of a target value. For example, unless otherwise specified in the explicit context of their use, the terms “substantially equal,” “about equal” and “approximately equal” mean that there is no more than incidental variation between among things so described. In the art, such variation is typically no more than $\pm 10\%$ of a predetermined target value.

It is to be understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments described herein are, for example, capable of operation in other orientations than those illustrated or otherwise described herein.

Unless otherwise specified the use of the ordinal adjectives “first,” “second,” and “third,” etc., to describe a common object, merely indicate that different instances of like objects are being referred to and are not intended to imply that the objects so described must be in a given sequence, either temporally, spatially, in ranking or in any other manner.

The terms “left,” “right,” “front,” “back,” “top,” “bottom,” “over,” “under,” and the like in the description and in the claims, if any, are used for descriptive purposes and not necessarily for describing permanent relative positions. For example, the terms “over,” “under,” “front side,” “back side,” “top,” “bottom,” “over,” “under,” and “on” as used herein refer to a relative position of one component, structure, or material with respect to other referenced components, structures or materials within a device, where such physical relationships are noteworthy. These terms are employed herein for descriptive purposes only and predominantly within the context of a device z-axis and therefore may be relative to an orientation of a device. Hence, a first material “over” a second material in the context of a figure provided herein may also be “under” the second material if the device is oriented upside-down relative to the context of the figure provided. In the context of materials, one material disposed over or under another may be directly in contact or may have one or more intervening materials. Moreover, one material disposed between two materials may be directly in contact with the two layers or may have one or more intervening layers. In contrast, a first material “on” a second material is in direct contact with that second material. Similar distinctions are to be made in the context of component assemblies.

The term “between” may be employed in the context of the z-axis, x-axis or y-axis of a device. A material that is between two other materials may be in contact with one or both of those materials, or it may be separated from both of the other two materials by one or more intervening materials. A material “between” two other materials may therefore be in contact with either of the other two materials, or it may be coupled to the other two materials through an intervening material. A device that is between two other devices may be directly connected to one or both of those devices, or it may be separated from both of the other two devices by one or more intervening devices.

As used throughout this description, and in the claims, a list of items joined by the term “at least one of” or “one or more of” can mean any combination of the listed terms. For example, the phrase “at least one of A, B or C” can mean A; B; C; A and B; A and C; B and C; or A, B and C. It is pointed out that those elements of a figure having the same reference numbers (or names) as the elements of any other figure can operate or function in any manner similar to that described, but are not limited to such.

In addition, the various elements of combinatorial logic and sequential logic discussed in the present disclosure may pertain both to physical structures (such as AND gates, OR gates, or XOR gates), or to synthesized or otherwise optimized collections of devices implementing the logical structures that are Boolean equivalents of the logic under discussion.

The terms “functional block,” “functional unit,” or “component” herein generally refer to any circuitry that performs a particular function. A “functional block” or “component” may be a unit of logic, circuit, cell, or chip layout that is reusable. A functional block is sometimes colloquially referred to as an IP (intellectual property) block. A few examples of functional blocks or components include processor cores, memories, caches, floating point processors, memory controllers, bus controllers, graphics processors, transceivers, network interface controllers, and display controllers. One or more portions of a larger functional block can themselves be designated as functional blocks. For example, an instruction execution unit and cache controller can be functional units or components of a processor functional unit. It should be appreciated that the foregoing examples are a non-exhaustive list of functional blocks.

As used herein, the terms “circuit” and “circuitry” comprise various electronic and electrical devices (“hardware”). Examples of hardware include analog circuits and analog circuit components (e.g., resistors, capacitors, inductors, diodes, and transistors). Other examples of hardware include digital circuits and digital circuit components, such as logic devices implementing Boolean functions. Examples of digital circuits include programmable logic devices (PLD), field programmable gate arrays (FPGA), application specific integrated circuits (ASIC), processors, processor cores, microprocessors, microcontrollers, digital signal processors (DSP), and graphics processing units (GPU). In yet another example, hardware includes a circuit that may be synthesized using a hardware description language (HDL) and which implements a state machine or other logic circuit. It should be understood that when hardware executes instructions stored in a memory device, the term hardware includes the stored instructions. Additional examples of hardware include volatile and non-volatile memory devices, such as registers, read-only memory (ROM), random access memory (RAM), and flash memory. Circuits and circuitry can include two or more instances of circuitry. Circuits and circuitry may comprise a combination of hardware elements

that cooperate to provide one or more functions. A particular instance of a circuits and circuitry may be referred to with a descriptive or non-descriptive label. For example, instances of circuits and circuitry that perform various functions may be referred to as receiver circuitry, processor circuitry, first circuit, or second circuit. Each of two or more instances of a circuit and circuitry can be comprised of distinct components. In addition, two or more instances of a circuit or circuitry can share one or more common components or resources.

As used herein, the term “hardware interface” refers to one or more physical components of a given device, where said one or more physical components accommodate coupling to interact with one or more physical components of another device. For example, a hardware interface may comprise conductive contacts of a connector or a socket that receives a connector. As another example, a hardware interface may comprise metal contacts, pads, metallization features, or other interconnect structures on a surface of or within a circuit board or integrated circuit (IC) chip. As a further example, a hardware interface may comprise an interconnect between contacts of respective components, such as solder or an interposer.

As used herein, the term “packaged device” means an integrated circuit (IC) comprising at least one IC chip. A packaged device may be a single IC chip or a host IC chip and one or more IC chiplets coupled to the host IC chip. The term “IC chiplet” refers to an IC die structure that is smaller than the host chip. A host die and an IC chiplet may be electrically interconnected with one another via a hardware.

FIG. 1 is a functional block diagram illustrating elements of a computing system to limit electric current consumption of a module that receives power from a bus in accordance with some embodiments.

Computing system **100** comprises a system-on-a-chip (SOC) **102**. In some embodiments, system **100** includes a processor in lieu of SoC **102**. System **100** further comprises module **104**, a bus **106**, a module circuit board **108**, and a system circuit board **110** (which may be referred to in the art as a mother board). All or part of bus **106** is attached to or an integral component of system circuit board **110**.

SOC **102** may include processor **112**, memory **114**, and first notifier circuitry **116**. It will be understood that SOC **102** includes additional components generally included in an SOC and not all components of SOC **102** are shown in FIG. 1. In embodiments in which a processor is provided in lieu of SOC **102**, various components typically included in a computer system may be provided as discrete components physically separate from the processor. Examples of components generally or typically included in an SOC or provided with a computer system include hardware and software for providing audio and display functions and input/output (I/O) functions, nonvolatile and volatile memory devices, and connectivity devices, such as a network interface, a cellular interface, or a wireless interface. In embodiments, SOC **102** may be the same as or similar to SOC **801**, processor **112** may be the same as or similar to processor **804**, and memory **114** may be the same as or similar to memory **830**. SOC **801**, processor **804**, and memory **830** are described below with reference to FIG. 8. As described below, hardware, software, or both for a particular function of computing system **100** may be provided by module **104**. In some embodiments, first notifier circuitry **116** may be a standalone component attached to system circuit board **110**.

According to various embodiments, bus **106** comprises a plurality of bus lines L_1 - L_N , e.g., wires or signal traces. In this description, a PCIe bus may be used as an example of

bus **106**. Similarly, module **104** and/or module circuit board **108** may be a PCIe module, which is used as an example of a device coupled to bus **106**. However, the present disclosure is not limited to PCIe devices or buses. It should be understood that descriptions about a PCIe bus or module may be applicable to any other device coupled to any computer bus, communication bus, or bus.

In various embodiments, bus **106** is a bus that conforms to or is compatible with an industry standard. As one example, bus **106** conforms to Peripheral Component Interconnect Express (PCI Express® or PCIe®) bus standard, e.g., PCI Express Base Specification Revision 6.0, version 1.0, Jan. 11, 2022, or any revision preceding Revision 6.0, ver. 1. PCI Express connects “modules” to a “host” with a “link.” A link can include one to 16 “lanes.” Each lane is composed of two differential signaling pairs. Two bus lines (one differential signaling pair) are used to receive data; the other two bus lines (the other differential signaling pair) are used to transmit data. Thus, each lane comprises four of bus lines L_1 - L_N . In the PCI Express specification, a host is an electrical source/master, e.g., a processor or SOC **102**, a “platform” is a physical location, such as system circuit board **110**, on which an add-in card or module, e.g., module **104**, are mounted, and a module is an add-in card that is either plugged into a connector, e.g., socket **120**, on the platform, or soldered on to the platform. An “add-in card” is a card or module plugged into a connector and mounted in a chassis socket.

The M.2 PCI Express specification defines interconnects based on a 75-position edge card connection scheme. The specification also defines an LGA (land grid array) pattern on the backside of a board that can be connected to a grid of contacts on a circuit board in a soldered-down scheme. In this description, a connector and a socket (or slot) may be used as an example of an edge card connection scheme. However, the present disclosure is not limited to edge card connection schemes. It should be understood that descriptions about edge card connection schemes may be applicable to a circuit board, e.g., module circuit board **108** or module **104**, in a soldered-down scheme.

While an M.2 connector has 75 positions, some of the positions are used for a connector key. For example, eight positions may be used for the connector key so that the number of pins or contacts that connect to bus lines is 67. According to one or more versions of the M.2 PCI Express specification, the maximum current rating per contact in a connector is 0.5 A (continuous). In this description, a current rating of 0.5 A is used as an example of a threshold current capacity. However, the present disclosure is not limited to this current rating. Any desired current rating may be used as a threshold current capacity.

A variety of module sizes and form factors are defined in the M.2 PCI Express specification. In addition, several distinct sockets are defined, e.g., Socket 1, Socket 2, Socket 3. A connectivity socket is designated Socket 1. It may be suitable for WiFi, BT, NFC, WiGig. A WWAN/SSD/Other socket is designated as Socket 2. It may be suitable for WWAN+GNSS solutions, and various SSD and SSD cache configurations. An SSD Drive socket is designated as Socket 3. Each socket type has its own bus line configuration and mechanical key. Socket 1 utilizes a single 3.3 V bus line or pins as a power source for a PCIe module. Socket 2 utilizes five 3.3 V bus lines or pins as a power source. Socket 3 utilizes nine 3.3 V bus lines or pins as a power source. Thus, 1, 5, or 9 of bus lines L_1 - L_N may be used to provide a power supply to a module, e.g., module **104**. In this description, Socket 2 may be used as an example. However, the present

disclosure is not limited to Socket 2. It should be understood that descriptions about Socket 2 may be applicable to embodiments in which Sockets 1 or 3 are employed, or in which an LGA pattern on the backside of a board, e.g., module board **108**, are connected to contacts of a motherboard, e.g., system circuit board **110**, in a soldered-down scheme. In addition, it should be understood that descriptions about Socket 2 may be applicable to embodiments employing sockets of any type and not only sockets defined by the M.2 PCIe standard.

In addition to specifying bus lines for supplying power to a PCIe module, the M.2 PCI Express specification defines uses of other bus lines. As one example, a Socket 2 pinout may include communication specific signals, PCI-e signals, USB (universal serial bus) signals, HSIO (high speed input output) signals, SSIC signals, SATA (serial advanced technology attachment) signals, UIM (user identity module) signals, and module configuration (CONFIG) pins. Various bus lines L_1 - L_N are used for these signals.

Socket 2 communication specific signals include W_DISABLE (wireless disable—disables radio operation on add-in cards that implement radio frequency applications) and COEX[0 . . . 3] (coexistence between WWAN and WiFi+BT on Socket 1). Various bus lines L_1 - L_N are used for these communication specific signals.

Socket 2 supplemental communication specific signals include FULL_CARD_POWER_OFF, RESET (resets module), GPIO[0 . . . 11] lines (general purpose input output—can be used for various functions), ANTCTL[0 . . . 3] (antenna control), IPC[0 . . . 7] (interprocess communication signals), AUDIO[0 . . . 3], WAKE_ON_WWAN (used to wake platform by WWAN device), and DPR (signal from SAR sensor). Various bus lines L_1 - L_N are used for these supplemental communication specific signals.

Module circuit board **108** (sometimes referred to in the art as an “add-in card” or “expansion card”) comprises a connector **118**. A socket **120** is connected to bus **106** and is adapted to receive connector **118**. In some embodiments, an LGA pattern on the backside of module board **108** may be provided in lieu of connector **118** and socket **120**, and module board **108** is coupled to bus **106** via contacts of system circuit board **110** in a soldered-down scheme.

A variety of circuit board sizes and form factors, as well as a variety of socket and connector types, are defined in the M.2 PCI Express specification. Module circuit board **108** and connector **118** may be any suitable size, form, or type, either in defined in the M.2 PCI Express specification or otherwise defined. Connector **118** may be mounted on or integral with module circuit board **108**. As one example, connector **118** may comprise a plurality of metal pads, pins, or other interconnect structures **122** on a surface of a connector at an edge of the surface. In an embodiment, connector **118** comprises 67 conductive pads or pins, and a connector key. In various embodiments, socket **120** (sometimes referred to in the art as a “slot”) comprises a plurality of conductive pads or surfaces **124** that correspond with the pads, pins, or other interconnect structures of connector **118**. In various embodiments, interconnect structures **122** of connector **118** are inserted into an opening in socket **120** where they contact conductive surfaces **124**.

Module **104** may be any type of module that provides a particular function of computing system **100**. In some embodiments, module **104** is a PCIe module. In various embodiments, module **104** comprises module circuitry **126**, memory **115**, first sampler circuitry **128**, utilization monitoring and power estimating circuitry **130**, and capability setting circuitry **132**.

Module circuitry **126** comprises hardware, software, firmware, or a combination of hardware, software, and firmware for performing one or more functions of computer system **100**. In addition, in embodiments, module circuitry **126** comprises circuitry to participate in a communication with processor **112** via connector **118** and bus **106**. In some embodiments, module **104** may be a graphics processing card and module circuitry **126** may comprise a GPU (graphics processing unit) and a memory. In some embodiments, module **104** may be a memory card and module circuitry **126** may comprise flash memory devices, a solid-state drive, or a hard disk drive. In some embodiments, module **104** may be a network card and module circuitry **126** may comprise a network interface controller. In some embodiments, module **104** may be a Wi-Fi® module, a Bluetooth® module, a satellite navigation module, a near field communication (NFC) module, a digital radio module (e.g., 2G, 3G, LTE), a WiGig® (60 GHz WiFi) module, a wireless wide-area network (WWAN) module, a wireless local-area network (WLAN) module, and module circuitry **126** may include circuitry needed to implement functions of the respective module.

Depending on the type of functions module **104** performs, module circuitry **126** may communicate any of a variety of types of signals with processor **112**, e.g., a data signal, an address signal, a control signal, or a supply (or other voltage) signal.

First sampler circuitry **128** may be connected to a particular one of the bus lines L_1 - L_N , e.g., L_{N-2} and to utilization monitoring and power estimating circuitry **130**. In various embodiments, first notifier circuitry **116** in SOC **102** is also connected to the particular one of the bus lines L_{N-2} . In embodiments, first notifier circuitry **116** places a signal on the particular one of the bus line L_{N-2} . The signal identifies a threshold current capacity of the connector **118**, socket **120**, or one or more bus lines. The first notifier circuitry **116** may read the threshold current capacity from a memory, e.g., memory **114**, and be based on a value in an industry standard or a data sheet of a manufacturer. In an embodiment, first sampler circuitry **128** samples the signal on the particular bus line L_{N-2} to detect the threshold current capacity. First sampler circuitry **128** may sample the bus line during a configuration or start-up stage of module **126**. Based on the sampled signal, first sampler circuitry **128** provides the threshold current capacity to utilization monitoring and power estimating circuitry **130**.

In embodiments, a threshold current capacity may be the threshold current capacity of a connector, socket, or bus line. For example, the threshold current capacity of a single connector, socket, or bus line may be a threshold peak current or a threshold average current of the applicable interconnect structure, conductive surface, or conductive trace. In embodiments, a threshold current capacity may be the threshold current capacity of a multiple connectors, sockets, or bus lines. For example, the threshold current capacity of multiple connectors, sockets, or bus lines may be a threshold peak current or a threshold average current of a single connector, socket, or bus line.

Utilization monitoring and power estimating circuitry **130** detects a utilization state of module **104** or module circuitry **126**. Based on the detected utilization state of module **104** or module circuitry **126**, utilization monitoring and power estimating circuitry **130** determines a level of current to be conducted within one or more of individual conductive interconnect structures **122**, conductive surfaces **124**, or bus lines. Utilization monitoring and power estimating circuitry **130** generates a signal which comprises an indication of

whether the level of current exceeds a threshold current capacity of the connector **118**, socket **120**, or bus **106**. As used herein and in the claims, the term “connector” may refer to any of connector **118**, socket **120**, or bus **106**.

The detected utilization state may be a utilization state in which module circuitry **126** is currently operating, e.g., an actual utilization state, or it may be a utilization state that module circuitry **126** has been requested to or expects to enter, e.g., an expected utilization state. A utilization state may be an explicitly defined state, e.g., idle mode or 3G connection transmission mode in a radio, or a state implied by a request to perform a particular function, e.g., a request to download or decompress a file of a particular size.

In embodiments, a utilization state may comprise one or more characteristics of operation by module circuitry **126**, or one or more characteristics of an environment in which module circuitry **126** operates. In some embodiments, a utilization state may comprise one or more actual characteristics. In some embodiments, a utilization state may comprise one or more predicted characteristics, e.g., a characteristic which is predicted based on a level of a metric, a rate of change of the metric, or both. A characteristic based on a rate of change may be a first order, second order, or other order rate of change. One example of an environmental characteristic is an availability of a network. Network availability refers to the strength and quality of a network signal. If signal strength is high and signal quality is good, a modem may switch to a higher order modulation scheme to enhance throughput and communication quality. As modulation schemes become more sophisticated, more modem resources may be used, and modem power consumption can increase. Conversely, when signal strength and quality is weak, a modem may switch to lower order modulation scheme to maintain link reliability. A less sophisticated modulation scheme may use fewer modem resources and require a low power level. Examples of operational characteristics include a quantity of resources used, such as an amount of memory, or a number of memory accesses or processing cycles in a time period. Another example of a characteristic of a utilization state is a measured value of a temperature of module circuitry **126** or another component. In an embodiment, temperature sensing circuitry may be provided in system **100**. Another example of a characteristic of a utilization state is throughput. In yet another example, the number of component carriers (CA), which are frequency blocks, that are used or are expected to be used may be a characteristic of a utilization state. In wireless communication, the data rate per user can be increased using a carrier aggregation technique, i.e., by increasing the number of component carriers assigned to the same modem. In a further example, a MIMO (multiple input and multiple output) radio communication technique may simultaneously send and receive two or more data signals over the same radio channel by utilizing multipath propagation. MIMO may employ multiple antennas. Various MIMO configurations may be characteristics of a utilization state.

In an embodiment, a utilization state includes or otherwise corresponds to a respective one or more conditions, each of a different respective metric of one or more power performance metrics. In an embodiment, a power performance metric is a peak current capacity. In an embodiment, a power performance metric is an average current capacity for a specified time period. In an embodiment, a power performance metric is the current conducted in a single interconnect structure **122** or conductive surface **124**. In an embodiment, a power performance metric is the sum of the individual currents conducted by each of two or more

interconnect structures **122** or conductive surfaces **124**. In other embodiments, a power performance metric is peak or average power (wattage) for a specified time period in one or more interconnect structures **122** or conductive surfaces **124**. In yet other embodiments, a power performance metric is a thermal dissipation value. Example thermal dissipation values for a WWAN module include: thermal dissipation of 4.8 W at a 105° die temp for an LTE cat. 4 150/50 Mbps use case; thermal dissipation of 6.0 W at a 105° die temp for an LTE cat. 16 980/150 Mbps use case; and thermal dissipation of 8.5 W at a 105° die temp for an ENDC: 5G NR (higher) Max DL+LTE (lower) MAX DL, where NR downlink 2.3 Gbps and LTE DL 1.2 Gbps, and uplink 150 Mbps use case. In the foregoing examples of thermal dissipation values, the use cases for the WWAN module include (a) concurrent video conference, plus content record, plus Office Productivity, plus download; and (b) Skype Desktop v8 two-way call, plus Office Productivity, plus download Skype record.

In various embodiments, utilization monitoring and power estimating circuitry **130** comprises a memory or circuitry that accesses a memory of module **104**, e.g., memory **115**, that stores reference data corresponding with performance profile information. Using the reference data, circuitry **130** may determine one or more power performance metrics. Reference data for each utilization state of module circuitry **126** specifies one or more characteristics of the utilization state, e.g., a level or a value, of a different power performance metric. For example, for a WWAN module, a characteristic of a first utilization state, e.g., one CA, may be a power performance metric of 1 W (or equivalent current), and a characteristic of a second utilization state, e.g., two CAs, may be a power performance metric of 2 W (or equivalent current). Module circuitry **126** may be operated in each utilization state under different environmental conditions and use cases to obtain performance profile information for the respective utilization states. The performance profile information may comprise one or more power performance metrics, e.g., peak or average power, peak or average current, for each utilization state.

Capability setting circuitry **132** may be coupled with utilization monitoring and power estimating circuitry **130** in order to receive the signal generated by circuitry **130** indicative of whether the current capacity threshold has been or is predicted to be exceeded. Capability setting circuitry **132** may also be coupled with module circuitry **126**. In embodiments, module circuitry **126** has multiple operational modes in which it can operate, and, based on the signal, capability setting circuitry **132** selects any one of the multiple operational modes. In an embodiment, capability setting circuitry **132** selects an operational mode associated with a power budget that does not require supply current exceeding a threshold current capacity of a connector, socket, or bus line. Capability setting circuitry **132** also configures module circuitry **126** to operate in the selected operational mode. In response to being configured by capability setting circuitry **132**, module circuitry **126** operates in the selected operational mode.

In embodiments, module circuitry **126** comprises WWAN capability and an operational mode of module circuitry **126** may be a particular number of component carrier connections or a particular transmit power value, e.g., as shown in FIG. 3. In embodiments, module **104** comprises a solid state drive (SSD) memory and operational modes of module circuitry **126** comprise PCIe dynamic power allocation (DPA) sub-states. DPA provides a capability to dynamically allocate power for a device. A DPA scheme can be used to

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specify a plurality of sub-states for a function, each sub-state defining a power allocation. For example, three sub-states may be specified for a function, wherein a first sub-state defines a 1 W power allocation, a second sub-state—defines a 2 W power allocation, and a third sub-state defines a 5 W power allocation. In embodiments, module **104** comprises a graphics processing module (GPU), and an operational mode is a particular clock value. GPU cores operate in synchrony with a clock. A GPU may be operated at multiple different clock speeds, each clock rate corresponding with one of multiple operational modes. While example operational modes are described with respect to specific module **126** functionality, e.g., WWAN, SSD, and GPU functions, it should be appreciated that, in embodiments, an operational mode for any type of module circuitry **126** may include a component carrier connection value, a transmit power value, a DPA sub-state, a clock speed, or any other suitable parameter.

FIG. 2 is a functional block diagram illustrating elements of a computing system to limit electric current consumption of a WWAN module that receives power from a bus in accordance with some embodiments.

Computing system **200** comprises a system-on-a-chip (SOC) **202**. In some embodiments, system **200** includes a processor in lieu of SOC **202**. System **200** further comprises WWAN module **204**, bus **206**, module circuit board **208**, and a system circuit board **210**. All or part of bus **206** is attached to or an integral component of system circuit board **210**.

SOC **202** may include processor **212**, memory **214**, first notifier circuitry **216**. It will be understood that SOC **202** includes additional components included in a SOC and not all components of SOC **202** are shown in FIG. 1. In embodiments in which a processor is provided in lieu of SOC **202**, various components typically included in a computer system may be provided as discrete components physically separate from the processor. In embodiments, SOC **202** may be the same as or similar to SOC **801**, processor **212** may be the same as or similar to processor **804**, and memory **214** may be the same as or similar to memory **830**. First notifier circuitry **216** may be the same as or similar to first notifier circuitry **116**.

Bus **206** comprises a plurality of bus lines, wires or signal traces. In various embodiments, bus **206** is a bus that conforms to or is compatible with an industry standard. In an embodiment, bus **106** conforms to PCI Express bus standard. Bus **206** comprises L power supply bus lines V_{SUPPLY} . Bus **206** comprises M interprocess communication bus lines, such as PCIe bus lines or USB bus lines. Bus **206** comprises N other bus lines, and one or more configuration bus lines, e.g., Config_pin_a and Config_pin_b.

WWAN module **204** may be an add-in card that is either plugged into a connector, e.g., a socket, or soldered on to system circuit board **210**. Hardware interface **218** connects various components on module circuit board **208** with bus **206**. In an embodiment, hardware interface **218** conforms to the M.2 PCI Express specification. In an embodiment, hardware interface **218** comprises a socket to receive a connector. In an embodiment, hardware interface **218** comprises contacts, pads, metallization features, or other interconnect structures on a surface of or within module circuit board **208** and corresponding contacts or other conductive features on a surface of system circuit board **210**.

WWAN module **204** comprises power management integrated circuit (PMIC) **220**, e.g., to implement various power management operations for module **204**. In addition, WWAN module **204** comprises various components to

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implement WWAN functions, such as baseband processor **222**, transceiver **224**, radio frequency (RF) front end **226**, and pull-up resistors R3, R4.

PMIC **220** receives power supply signals on bus lines L. PMIC **220** provides power ($V_{CORE SUPPLY}$) needed to operate baseband processor **222**. PMIC **220** also provides power ($V_{I/O SUPPLY}$) needed to operate input/output components of baseband processor **222**.

In an embodiment, WWAN module **204** comprises sampler circuitry **228**, which may be the same as or similar to second sampler circuitry **128**. In an embodiment, WWAN module **204** comprises utilization monitoring and power estimating circuitry **230**, which may be the same as or similar to utilization monitoring and power estimating circuitry **130**. In an embodiment, WWAN module **204** comprises capability setting circuitry **232**, which may be the same as or similar to capability setting circuitry **132**. In an embodiment, sampler circuitry **228**, utilization monitoring and power estimating circuitry **230**, and capability setting circuitry **132** may be within baseband processor **222**, however, this is not essential.

Sampler circuitry **228** within baseband processor **222** may be coupled to Config_pin_a. First notifier circuitry **216** places a signal on Config_pin_a that identifies a threshold current capacity of hardware interface **218**, e.g., a connector or socket, or both. Sampler circuitry **228** of baseband processor **222** samples the signal on Config_pin_a to detect the threshold current capacity. The signal may be sampled during a configuration or start-up stage of module **204**. Based on the sampled signal, utilization monitoring and power estimating circuitry **230** detects an actual or expected utilization state of baseband processor **222** and/or other components of WWAN module **204**.

Based on the detected utilization state, utilization monitoring and power estimating circuitry **230** determines a level of current to be conducted with hardware interface **218** or a connector, socket, or other interconnect structure of bus **206**. Utilization monitoring and power estimating circuitry **230** generates a signal which comprises an indication of whether the level of current exceeds a threshold current capacity of the connector, socket, or other interconnect.

WWAN module **204** has multiple operational modes in which it can operate, and, based on the signal, capability setting circuitry **232** selects any one of the multiple operational modes. Capability setting circuitry **232** also configures WWAN module **204** to operate in the selected operation mode. In response to being configured by capability setting circuitry **232**, WWAN module **204** operates in the selected operation mode. In an embodiment, the multiple operational modes of WWAN module **204** comprise operational modes of baseband processor **222**.

FIG. 3 illustrates a state diagram **300** for a baseband processor, e.g., baseband processor **222**, in accordance with some embodiments. State diagram **300** illustrates multiple operational modes for a baseband processor. Each operational mode may be associated with a predicted power level and (by implication) current level. The operational states include at least one RAT (radio access technology) idle operational mode **302**, which is associated with a predicted power level of 1 W. In operational mode **304**, the baseband processor uses 2G connection technology, which is associated with a predicted power level of 3 W. In operational mode **306**, the baseband processor uses 3G connection technology, which is associated with a predicted power level of 3 W. In operational mode **308**, the baseband processor uses LTE (long-term evolution) connection technology. Operational mode **308** may comprise a plurality of opera-

tional modes, depending on a transmit power P_{TX} used and a number of CCs (component carriers) used. For example, operational mode 310 corresponds with two component carriers and a third level of transmit power, and is associated with a predicted power level of 5 W. As another example, operational mode 312 corresponds with three component carriers and a first level of transmit power, and is associated with a predicted power level of 2 W. It will be understood by one of ordinary skill that once a number of metal pads, pins, or other interconnect structures of a connector or socket is specified, a predicted power level may be converted into a predicted level of current in each of the pads, pins, or other interconnect structures.

FIG. 4 illustrates a flow diagram of a process 400 for limiting current in a module according to various embodiments. In various embodiments, process 400 may be employed with any type of PCIe module. In various embodiments, process 400 may be employed with an add-in card having a connector plugged into a socket. In an embodiment, process 400 may be employed with any type of WWAN module.

At 402, a threshold current capacity of a connector or socket is received. The threshold current or power capacity may be received during a configuration or start-up stage of module. The threshold current capacity or power may be received by sampler circuitry that samples a signal on a particular bus line, e.g., a configuration pin.

At 404, an actual or expected utilization state of multiple utilization states of the module is detected. Each utilization state may comprise one or more characteristics of operation (e.g., levels/values) of the module. Each characteristic may comprise one or more power performance metrics. The utilization state may be detected by utilization monitoring and power estimating circuitry.

At 406, a power budget is determined. The power budget is an estimate of power that will be used or a rate of current that will be required in the actual or expected utilization state. The power budget may be determined using reference data corresponding with a plurality of utilization states, which may be stored in a memory. The power budget may be determined by utilization monitoring and power estimating circuitry.

At 408, the power budget is used to determine whether a current capacity threshold is predicted to be exceeded. The threshold current capacity may be of a connector, socket, or other interconnect. The determination of whether a current capacity threshold may be exceeded may be performed by utilization monitoring and power estimating circuitry.

If current capacity threshold may be exceeded, the module may be configured to operate in a first operational mode at 410. When operating in the first operational mode, the module does not or is not expected to exceed the current capacity threshold. In some circumstances, an operational mode in which the module operates at or below the current capacity is not available. In these circumstances, the module may be shut down at 412. The configuring of the module in the first operational mode 410 or the shutting down of the module at 412 may be performed by capability setting circuitry.

Subsequent to the module being configured to operate in the first operational mode at 410 or it being determined that the current capacity threshold is not predicted to be exceeded at 408, the module is operated at 414.

FIG. 5 is a functional block diagram illustrating elements of a computing system to assure that the voltage level used by a module and by a host for sideband signaling match in accordance with some embodiments.

Computing system 500 comprises a system-on-a-chip (SOC) 502. In some embodiments, system 500 includes a processor in lieu of SOC 502. System 500 further comprises module 504, a bus 506, a module circuit board 508, and a system circuit board 510. All or part of bus 506 may be attached to or an integral component of system circuit board 510.

SOC 502 may include processor 512, memory 514, second sampler circuitry 517, and level shifting control circuitry 534. It will be understood that SOC 502 includes additional components included in an SOC and not all components of SOC 502 are shown in FIG. 5. In some embodiments a processor is provided in lieu of SOC 502; in these embodiments, various components typically included in a computer system may be provided as discrete components physically separate from the processor. Examples of components typically included in a SOC or provided with a computer system include hardware and software for providing audio and display functions and input/output (I/O) functions, nonvolatile and volatile memory devices, and connectivity devices, such as a network interface, a cellular interface, or a wireless interface. In embodiments, SOC 502 may be the same as or similar to SOC 801, processor 512 may be the same as or similar to processor 804, and memory 514 may be the same as or similar to memory 830.

According to various embodiments, bus 506 comprises a plurality of bus lines L_1 - L_N , e.g., wires or signal traces. In various embodiments, bus 506 is a bus that conforms to or is compatible with an industry standard. As one example, bus 506 conforms to the PCIe bus standard. Bus 506 may be the same as or similar to bus 106. It is not essential that bus 506 be a PCIe bus. In various embodiments, bus 506 may be any computer bus, communication bus, or bus.

In various embodiments, one or more of bus lines L_1 - L_N of bus 506 may be used for sideband signals. As used herein, the term "sideband signal" comprises FULL_CARD_POWER_OFF, RESET, W_DISABLE, and COEX[0 . . . 3], and CONFIG[0 . . . 3]. While these signals are used as examples of sideband signals in this description, it should be appreciated that, in some embodiments, sideband signals may include other signals, such as any of the signals defined in an industry standard, e.g., PCI Express, or otherwise deemed necessary for a particular design.

Module circuit board 508 comprises a connector 518. A socket 520 is connected to bus 506 and is adapted to receive connector 518. Connector 518 may be mounted on or integral with circuit board 508. As one example, connector 518 may comprise a plurality of metal pads, pins, or other interconnect structures 522 on a surface of a connector at an edge of the surface. In an embodiment, connector 518 comprises 67 conductive pads or pins, and a connector key. In various embodiments, socket 520 comprises a plurality of conductive pads or surfaces 524 that correspond with the pads or pins of connector 518. In various embodiments, conductive pads or pins 522 of connector 518 are inserted into an opening in socket 520 where they contact conductive surfaces 524. In various embodiments, connector 518 may be similar to or the same as connector 118. In various embodiments, socket 520 may be similar to or the same as socket 120. In some embodiments, an LGA pattern on the backside of module board 508 may be provided in lieu of connector 518 and socket 520, and module board 508 is coupled to bus 506 via contacts of system circuit board 510 in a soldered-down scheme.

Module 504 may be any type of module that provides a particular function of computing system 500. In various embodiments, module 504 comprises module circuitry 526,

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second notifier circuitry **536**, voltage regulator **507**, and EEPROM **540**. In some embodiments, module **504** is a PCIe module. In some embodiments, module **504** may be a Wi-Fi® module, a Bluetooth® module, a satellite navigation module, a near field communication (NFC) module, a digital radio module (e.g., 2G, 3G, LTE), a WiGig® (60 GHz WiFi) module, a WWAN module, a WLAN module, and module circuitry **126** may include circuitry needed to implement functions of the respective module.

Module circuitry **526** comprises hardware, software, firmware, or a combination of hardware, software, and firmware for performing one or more functions of computer system **500**. In addition, module circuitry **526** comprises circuitry to participate in a communication with processor **512** via connector **518**, socket **520**, and bus **506**. In some embodiments, module circuitry **526** may be similar to or the same as module circuitry **126**. Depending on the type functions module **504** performs, module circuitry **526** may communicate any of a variety of types of signals with processor **512**, e.g., a data signal, an address signal, a control signal, or a supply (or other voltage) signal. In embodiments, module circuitry **526** may communicate a sideband signal.

In various embodiments, module circuit board **508** includes second notifier circuitry **536**, which may be coupled to module circuit board **508**, or included in module **504** as shown in the figure. In various embodiments, second notifier circuitry **536** participates in a communication with second sampler circuitry **517** via connector **518**, socket **520**, and one or more of bus lines L_1 - L_N of bus **506**. Communication by second notifier circuitry **536** with second sampler circuitry **517** may be distinct from other communications that module circuitry **526** may have with processor **512**. In embodiments, second notifier circuitry **536** communicates a signal indicative of a sideband signaling voltage level or a voltage domain.

In various embodiments, SOC **502** includes second sampler circuitry **517**, which may be coupled to one of the bus lines L_1 - L_N , e.g., L_N . In some embodiments, second sampler circuitry **517** may be a standalone component attached to system circuit board **510**. Second sampler circuitry **517** communicates with second notifier circuitry **536** to detect a sideband signaling voltage level used by module **504** or module circuitry **526**. The sideband signaling voltage may be the same as a voltage domain of module **504** or module circuitry **526**. As used herein, a “voltage domain” is a group of functional circuitry that are operated at a same voltage provided to the group by a supply rail of a voltage regulator. Based on a result of the communication, second sampler circuitry **517** provides the sideband signaling voltage level used by module **504** or module circuitry **526** to level shifting control circuitry **534**. In an embodiment, second sampler circuitry **517** provides a voltage domain used by module **504** or module circuitry **526** to level shifting control circuitry **534**.

In an embodiment, a ground pin of connector **518** which is not a return for a HSIO (high speed input output) bus line may be used for communications between second notifier circuitry **536** and second sampler circuitry **517**. For example, a ground pin of connector **518** may be used to communicate a sideband signaling voltage level or a voltage domain value.

In another embodiment, two or more pins of connector **518** not used for other purposes, e.g., I2C or GPIO lines, may be used for communications between second notifier circuitry **536** and second sampler circuitry **517**. For example, I2C or GPIO lines of connector **518** may be used to communicate a sideband signaling voltage level or a

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voltage domain value. In some embodiments, data or a signal indicative of a sideband signaling voltage level or a voltage domain may be defined as a hardware strap, e.g., as a bus line tied to a particular voltage, or a latch, switch, or jumper. In other embodiments, the signal indicative of data or a signal indicative of a sideband signaling voltage level or a voltage domain may be stored in a memory, e.g., EEPROM **540**. I2C or GPIO lines may be used to read a value stored in EEPROM **540**.

Second sampler circuitry **517** may sample the particular bus line when it detects the presence of module circuit board **508**. In some embodiments, second sampler circuitry **517** may sample the particular bus line during a configuration or start-up stage of module circuitry **526** or module **504**. In some embodiments, second sampler circuitry **517** reads data indicative of a sideband signaling voltage level from a hardware strap or a memory, e.g., EEPROM **540** when it detects the presence of module circuit board **508**, or at a start-up stage.

Module **504** or module circuitry **526** may use any known sideband signaling voltage level or voltage domain. For example, the signal may indicate that one of 1.2V, 1.6V, 1.8V, or 3.3V correspond with a logic one. The signal may indicate that all or part of module **504** or module circuitry **526** operates in one of 1.2V, 1.6V, 1.8V, or 3.3V voltage domains.

In some embodiments, level shifting control circuitry **534** controls one or more voltage level shifters or translators, e.g., VLT **538**, **539**. On receipt of the sideband signaling voltage level used by module **504** or module circuitry **526** from second sampler circuitry **517**, level shifting control circuitry **534** controls voltage level shifter **538** or **539** so that sideband signals transmitted between SOC **502** and module **504** are compatible with a respective voltage domain of SOC **502**, module **504**, or module circuitry **526**. In embodiments, voltage level shifter **538** may be a bidirectional level shifter. It will be appreciated that a voltage level shifter **539** is an optional alternative to voltage level shifter **538**. In various embodiment, one of voltage level shifters **538**, **539** may be provided for each line of bus **506** used for sideband signaling. In embodiments, voltage level shifter **538** may be a discrete component attached to system circuit board **510** or integral with SOC **502**. Voltage level shifter **538** may be connected to voltage regulator **509**, which may be attached to system circuit board **510**. In alternative embodiments, voltage level shifter **539** may be a discrete component attached to module circuit board **508** or integral with module **504**. Voltage level shifter **539** may be connected to voltage regulator **507**, which may be attached to module circuit board **508** or integral with module **504**. In addition to controlling the operation of voltage level shifters **538** or **539**, in some embodiments, level shifting control circuitry **534** may block module **504** from operating, e.g., when a required voltage level shifter is not present on a bus line in system **500**.

FIG. 5 discloses techniques in which second sampler circuitry **517** and level shifting control circuitry **534** of SOC **502** perform certain operations and second notifier circuitry **536** of module **504** performs other operations. It should be appreciated that other arrangements are contemplated. In some embodiments, circuitry of module circuit board **508** may detect a voltage domain of SOC **502** or processor **512**, thereby performing a sampling operation similar to that of second sampler circuitry **517**. In addition, circuitry (not shown) of module circuit board **508** may control a voltage level shifter **538**, **539** so that sideband signaling voltages are compatible with a voltage domain of circuitry external to

module circuit board **508**, e.g., SOC **502** or processor **512**, thereby performing an operation similar to that of level shifting control circuitry **534**. Furthermore, circuitry of SOC **502** or processor **512** may communicate a signal indicative of a sideband signaling voltage level or a voltage domain, thereby performing an operation similar to second notifier circuitry **536**.

FIG. 6 is a block diagram of voltage level shifter circuitry **600** in accordance with some embodiments. Voltage level shifter circuitry **600** comprises N-channel enhancement mode MOSFET **602**, and resistors **R1** and **R2**. Body diode **604** between drain and substrate may be internal to MOSFET **602** or may be connected externally. Resistor **R2** is connected to a supply voltage $V_{I/O\ MODULE}$ of module **504**. Resistor **R1** is connected to a supply voltage $V_{I/O\ HOST}$ of SOC **502**. In various embodiments, level shifting control circuitry **534** sets supply voltage $V_{I/O\ HOST}$ to be substantially equal to supply voltage $V_{I/O\ MODULE}$, thereby assuring that the voltage level used by the module **504** for sideband signaling will be compatible with voltage level used by SOC **502**.

Voltage level shifter circuitry **600** may be provided for one or more of bus lines L_1 - L_N of bus **506** used for sideband signaling, e.g., bus line L_Z . Voltage level shifter circuitry **600** has a bi-directional capability, i.e., it is capable of translating a sideband signal received from a module to a voltage that is compatible with a host and a sideband signal transmitted from the host to a voltage that is compatible with the module. In various embodiments, voltage level shifters **VLT 538, 539** comprise voltage level shifter circuitry **600**. While voltage level shifter circuitry **600** illustrates one technique for translating a signal from one voltage domain to another, other techniques are known in the art. It should be appreciated that embodiments of voltage level shifters **VLT 538, 539** are not limited to the method illustrated in FIG. 6. In other embodiments, any suitable technique for translating a signal from one voltage domain to another may be employed.

FIG. 7 illustrates a flow diagram of a process **700** for assuring that the voltage level used by a module for sideband signaling will be compatible with voltage level used by a host according to various embodiments.

At **702**, a host, e.g., an SOC or a processor, is powered up. At **704**, power on process is started for a module, such as an add-in board or module **108, 208**, or **508**. At **706**, the host detects a sideband signaling voltage level used by the module or a component of the module. The host may detect the sideband signaling voltage level with second sampler circuitry that communicates with second notifier circuitry in the module. The second sampler circuitry may sample a signal on a bus line of a bus, e.g., bus **106, 206**, or **506**, connecting the host and the module. Alternatively, the second sampler circuitry may read data stored in a memory of the module using bus lines of a bus, e.g., bus **106, 206**, or **506**, connecting the host and the module.

At **708**, it is determined whether the system includes voltage level shifter circuitry for translating a signal from one voltage domain to another. If the system includes voltage level shifter circuitry and the respective voltage domains are not compatible, the voltage level shifter circuitry is controlled at **710** so that sideband signals received from the module are translated to a voltage that is compatible with the host and sideband signals transmitted from the host are translated to a voltage that is compatible with the module. In addition, the module may be fully powered up at **710**. If the system does not include voltage level shifter circuitry at **712**, it is determined whether the respective

voltage domains of the host and module are compatible, e.g., the same. If the respective voltage domains are not compatible, powering up of the module is aborted and is disabled from being powered up at **714**. In addition, an error message may be sent to a user interface. If the respective voltage domains are compatible, the powering up process is completed at **716**. At **718**, an initialization process for the module is completed.

Elements of embodiments (e.g., flowchart with reference to FIGS. 3 and 7) are also provided as a machine-readable medium for storing the computer-executable instructions (e.g., instructions to implement any other processes discussed herein). In some embodiments, a computing platform comprises memory, a processor, machine-readable storage media (also referred to as tangible machine-readable medium), a communication interface (e.g., wireless or wired interface), and a network bus coupled together.

In some embodiments, the various logic blocks are coupled together via a Network Bus. Any suitable protocol may be used to implement the network bus. In some embodiments, machine-readable storage medium includes instructions (also referred to as the program software code/instructions) for calculating or measuring distance and relative orientation of a device with reference to another device as described with reference to various embodiments and flowchart.

Program software code/instructions associated with the flow diagram of FIGS. 3 and 7 (and/or various embodiments) and executed to implement embodiments of the disclosed subject matter may be implemented as part of an operating system or a specific application, component, program, object, module, routine, or other sequence of instructions or organization of sequences of instructions referred to as Pcode, "program software code/instructions," "operating system program software code/instructions," "application program software code/instructions," or simply "software" or firmware embedded in processor. In some embodiments, the program software code/instructions associated with the sequence flow diagrams of FIGS. 3 and 7 (and/or various embodiments) are executed by the system or one or more components thereof.

In some embodiments, the program software code/instructions associated with reference to FIGS. 3 and 7 (and/or various embodiments) are stored in a computer executable storage medium and executed by the processor. Here, computer executable storage medium is a tangible machine-readable medium that can be used to store program software code/instructions and data that, when executed by a computing device, causes one or more processors to perform a method(s) as may be recited in one or more accompanying claims directed to the disclosed subject matter.

The tangible machine-readable medium may include storage of the executable software program code/instructions and data in various tangible locations, including for example ROM, volatile RAM, non-volatile memory and/or cache and/or other tangible memory as referenced in the present application. Portions of this program software code/instructions and/or data may be stored in any one of these storage and memory devices. Further, the program software code/instructions can be obtained from other storage, including, e.g., through centralized servers or peer to peer networks and the like, including the Internet. Different portions of the software program code/instructions and data can be obtained at different times and in different communication sessions or in the same communication session.

The software program code/instructions (associated with reference to FIGS. 3 and 7 and other embodiments) and data

can be obtained in their entirety prior to the execution of a respective software program or application by the computing device. Alternatively, portions of the software program code/instructions and data can be obtained dynamically, e.g., just in time, when needed for execution. Alternatively, some combination of these ways of obtaining the software program code/instructions and data may occur, e.g., for different applications, components, programs, objects, modules, routines or other sequences of instructions or organization of sequences of instructions, by way of example. Thus, it is not required that the data and instructions be on a tangible machine readable medium in entirety at a particular instance of time.

Examples of tangible computer-readable media include but are not limited to recordable and non-recordable type media such as volatile and non-volatile memory devices, read only memory (ROM), random access memory (RAM), flash memory devices, floppy and other removable disks, magnetic storage media, optical storage media (e.g., Compact Disk Read-Only Memory (CD ROMS), Digital Versatile Disks (DVDs), etc.), ferroelectric memory, resistive RAM, phase change memory (PCM), magnetic RAM (MRAM, among others. The software program code/instructions may be temporarily stored in digital tangible communication links while implementing electrical, optical, acoustical or other forms of propagating signals, such as carrier waves, infrared signals, digital signals, etc. through such tangible communication links.

In general, tangible machine readable medium includes any tangible mechanism that provides (i.e., stores and/or transmits in digital form, e.g., data packets) information in a form accessible by a machine (i.e., a computing device), which may be included, e.g., in a communication device, a computing device, a network device, a personal digital assistant, a manufacturing tool, a mobile communication device, whether or not able to download and run applications and subsidized applications from the communication network, such as the Internet, e.g., an iPhone®, Galaxy®, or the like, or any other device including a computing device. In one embodiment, processor-based system is in a form of or included within a PDA (personal digital assistant), a cellular phone, a notebook computer, a tablet, a game console, a set top box, an embedded system, a TV (television), a personal desktop computer, etc. Alternatively, the traditional communication applications and subsidized application(s) may be used in some embodiments of the disclosed subject matter.

FIG. 8 illustrates a computer system or computing device **800** (also referred to as device **800**) to limit electric current consumption of a module that receives power from a bus, and to assure that the voltage level used by the module and by a host for sideband signaling match, in accordance with some embodiments. It is pointed out that those elements of FIG. 8 having the same reference numbers (or names) as the elements of any other figure can operate or function in any manner similar to that described, but are not limited to such.

In some embodiments, device **800** represents an appropriate computing device, such as a computing tablet, a mobile phone or smart-phone, a laptop, a desktop, an Internet-of-Things (IOT) device, a server, a wearable device, a set-top box, a wireless-enabled e-reader, or the like. It will be understood that certain components are shown generally, and not all components of such a device are shown in device **800**.

In an example, the device **800** comprises a SOC (System-on-Chip) **801**. An example boundary of the SOC **801** is illustrated using dotted lines in FIG. 8, with some example

components being illustrated to be included within SOC **801**—however, SOC **801** may include any appropriate components of device **800**.

In some embodiments, device **800** includes processor **804**. Processor **804** can include one or more physical devices, such as microprocessors, application processors, microcontrollers, programmable logic devices, processing cores, or other processing means. The processing operations performed by processor **804** include the execution of an operating platform or operating system on which applications and/or device functions are executed. The processing operations include operations related to I/O (input/output) with a human user or with other devices, operations related to power management, operations related to connecting computing device **800** to another device, and/or the like. The processing operations may also include operations related to audio I/O and/or display I/O.

In some embodiments, processor **804** includes multiple processing cores (also referred to as cores) **808a**, **808b**, **808c**. Although merely three cores **808a**, **808b**, **808c** are illustrated in FIG. 8, the processor **804** may include any other appropriate number of processing cores, e.g., tens, or even hundreds of processing cores. Processor cores **808a**, **808b**, **808c** may be implemented on a single integrated circuit (IC) chip. Moreover, the chip may include one or more shared and/or private caches, buses or interconnections, graphics and/or memory controllers, or other components.

In some embodiments, processor **804** includes cache **806**. In an example, sections of cache **806** may be dedicated to individual cores **808** (e.g., a first section of cache **806** dedicated to core **808a**, a second section of cache **806** dedicated to core **808b**, and so on). In an example, one or more sections of cache **806** may be shared among two or more of cores **808**. Cache **806** may be split in different levels, e.g., level 1 (L1) cache, level 2 (L2) cache, level 3 (L3) cache, etc.

In some embodiments, a given processor core (e.g., core **808a**) may include a fetch unit to fetch instructions (including instructions with conditional branches) for execution by the core **808a**. The instructions may be fetched from any storage devices such as the memory **830**. Processor core **808a** may also include a decode unit to decode the fetched instruction. For example, the decode unit may decode the fetched instruction into a plurality of micro-operations. Processor core **808a** may include a schedule unit to perform various operations associated with storing decoded instructions. For example, the schedule unit may hold data from the decode unit until the instructions are ready for dispatch, e.g., until all source values of a decoded instruction become available. In one embodiment, the schedule unit may schedule and/or issue (or dispatch) decoded instructions to an execution unit for execution.

The execution unit may execute the dispatched instructions after they are decoded (e.g., by the decode unit) and dispatched (e.g., by the schedule unit). In an embodiment, the execution unit may include more than one execution unit (such as an imaging computational unit, a graphics computational unit, a general-purpose computational unit, etc.). The execution unit may also perform various arithmetic operations such as addition, subtraction, multiplication, and/or division, and may include one or more an arithmetic logic units (ALUs). In an embodiment, a co-processor (not shown) may perform various arithmetic operations in conjunction with the execution unit.

Further, an execution unit may execute instructions out-of-order. Hence, processor core **808a** (for example) may be

an out-of-order processor core in one embodiment. Processor core **808a** may also include a retirement unit. The retirement unit may retire executed instructions after they are committed. In an embodiment, retirement of the executed instructions may result in processor state being committed from the execution of the instructions, physical registers used by the instructions being de-allocated, etc. The processor core **808a** may also include a bus unit to enable communication between components of the processor core **808a** and other components via one or more buses. Processor core **808a** may also include one or more registers to store data accessed by various components of the core **808a** (such as values related to assigned app priorities and/or sub-system states (modes) association).

In some embodiments, device **800** comprises connectivity circuitries **831**. For example, connectivity circuitries **831** includes hardware devices (e.g., wireless and/or wired connectors and communication hardware) and/or software components (e.g., drivers, protocol stacks), e.g., to enable device **800** to communicate with external devices. Device **800** may be separate from the external devices, such as other computing devices, wireless access points or base stations, etc.

In an example, connectivity circuitries **831** may include multiple different types of connectivity. To generalize, the connectivity circuitries **831** may include cellular connectivity circuitries, wireless connectivity circuitries, etc. Cellular connectivity circuitries of connectivity circuitries **831** refers generally to cellular network connectivity provided by wireless carriers, such as provided via GSM (global system for mobile communications) or variations or derivatives, CDMA (code division multiple access) or variations or derivatives, TDM (time division multiplexing) or variations or derivatives, 3rd Generation Partnership Project (3GPP) Universal Mobile Telecommunications Systems (UMTS) system or variations or derivatives, 3GPP Long-Term Evolution (LTE) system or variations or derivatives, 3GPP LTE-Advanced (LTE-A) system or variations or derivatives, Fifth Generation (5G) wireless system or variations or derivatives, 5G mobile networks system or variations or derivatives, 5G New Radio (NR) system or variations or derivatives, or other cellular service standards. Wireless connectivity circuitries (or wireless interface) of the connectivity circuitries **831** refers to wireless connectivity that is not cellular, and can include personal area networks (such as Bluetooth, Near Field, etc.), local area networks (such as Wi-Fi), and/or wide area networks (such as WiMax), and/or other wireless communication. In an example, connectivity circuitries **831** may include a network interface, such as a wired or wireless interface, e.g., so that a system embodiment may be incorporated into a wireless device, for example, cell phone or personal digital assistant.

In some embodiments, device **800** comprises control hub **832**, which represents hardware devices and/or software components related to interaction with one or more I/O devices. For example, processor **804** may communicate with one or more of display **822**, one or more peripheral devices **824**, storage devices **828**, one or more other external devices **829**, etc., via control hub **832**. Control hub **832** may be a chipset, a Platform Control Hub (PCH), and/or the like.

For example, control hub **832** illustrates one or more connection points for additional devices that connect to device **800**, e.g., through which a user might interact with the system. For example, devices (e.g., devices **829**) that can be attached to device **800** include microphone devices, speaker or stereo systems, audio devices, video systems or

other display devices, keyboard or keypad devices, or other I/O devices for use with specific applications such as card readers or other devices.

As mentioned above, control hub **832** can interact with audio devices, display **822**, etc. For example, input through a microphone or other audio device can provide input or commands for one or more applications or functions of device **800**. Additionally, audio output can be provided instead of, or in addition to display output. In another example, if display **822** includes a touch screen, display **822** also acts as an input device, which can be at least partially managed by control hub **832**. There can also be additional buttons or switches on computing device **800** to provide I/O functions managed by control hub **832**. In one embodiment, control hub **832** manages devices such as accelerometers, cameras, light sensors or other environmental sensors, or other hardware that can be included in device **800**. The input can be part of direct user interaction, as well as providing environmental input to the system to influence its operations (such as filtering for noise, adjusting displays for brightness detection, applying a flash for a camera, or other features).

In some embodiments, control hub **832** may couple to various devices using any appropriate communication protocol, e.g., PCIe (Peripheral Component Interconnect Express), USB (Universal Serial Bus), Thunderbolt, High Definition Multimedia Interface (HDMI), Firewire, etc.

In some embodiments, display **822** represents hardware (e.g., display devices) and software (e.g., drivers) components that provide a visual and/or tactile display for a user to interact with device **800**. Display **822** may include a display interface, a display screen, and/or hardware device used to provide a display to a user. In some embodiments, display **822** includes a touch screen (or touch pad) device that provides both output and input to a user. In an example, display **822** may communicate directly with the processor **804**. Display **822** can be one or more of an internal display device, as in a mobile electronic device or a laptop device or an external display device attached via a display interface (e.g., DisplayPort, etc.). In one embodiment display **822** can be a head mounted display (HMD) such as a stereoscopic display device for use in virtual reality (VR) applications or augmented reality (AR) applications.

In some embodiments and although not illustrated in the figure, in addition to (or instead of) processor **804**, device **800** may include Graphics Processing Unit (GPU) comprising one or more graphics processing cores, which may control one or more aspects of displaying contents on display **822**.

Control hub **832** (or platform controller hub) may include hardware interfaces and connectors, as well as software components (e.g., drivers, protocol stacks) to make peripheral connections, e.g., to peripheral devices **824**.

It will be understood that device **800** could both be a peripheral device to other computing devices, as well as have peripheral devices connected to it. Device **800** may have a “docking” connector to connect to other computing devices for purposes such as managing (e.g., downloading and/or uploading, changing, synchronizing) content on device **800**. Additionally, a docking connector can allow device **800** to connect to certain peripherals that allow computing device **800** to control content output, for example, to audiovisual or other systems.

In addition to a proprietary docking connector or other proprietary connection hardware, device **800** can make peripheral connections via common or standards-based connectors. Common types can include a Universal Serial Bus (USB) connector (which can include any of a number of

different hardware interfaces), DisplayPort including MiniDisplayPort (MDP), High Definition Multimedia Interface (HDMI), Firewire, or other types.

In some embodiments, connectivity circuitries **831** may be coupled to control hub **832**, e.g., in addition to, or instead of, being coupled directly to the processor **804**. In some embodiments, display **822** may be coupled to control hub **832**, e.g., in addition to, or instead of, being coupled directly to processor **804**.

In some embodiments, device **800** comprises memory **830** coupled to processor **804** via memory interface **834**. Memory **830** includes memory devices for storing information in device **800**. Memory can include nonvolatile (state does not change if power to the memory device is interrupted) and/or volatile (state is indeterminate if power to the memory device is interrupted) memory devices. Memory device **830** can be a dynamic random access memory (DRAM) device, a static random access memory (SRAM) device, flash memory device, phase-change memory device, or some other memory device having suitable performance to serve as process memory. In one embodiment, memory **830** can operate as system memory for device **800**, to store data and instructions for use when the one or more processors **804** executes an application or process. Memory **830** can store application data, user data, music, photos, documents, or other data, as well as system data (whether long-term or temporary) related to the execution of the applications and functions of device **800**.

Elements of various embodiments and examples are also provided as a machine-readable medium (e.g., memory **830**) for storing the computer-executable instructions (e.g., instructions to implement any other processes discussed herein). The machine-readable medium (e.g., memory **830**) may include, but is not limited to, flash memory, optical disks, CD-ROMs, DVD ROMs, RAMs, EPROMs, EEPROMs, magnetic or optical cards, phase change memory (PCM), or other types of machine-readable media suitable for storing electronic or computer-executable instructions. For example, embodiments of the disclosure may be downloaded as a computer program (e.g., BIOS) which may be transferred from a remote computer (e.g., a server) to a requesting computer (e.g., a client) by way of data signals via a communication link (e.g., a modem or network connection).

In some embodiments, device **800** comprises temperature measurement circuitries **840**, e.g., for measuring temperature of various components of device **800**. In an example, temperature measurement circuitries **840** may be embedded, or coupled or attached to various components, whose temperature are to be measured and monitored. For example, temperature measurement circuitries **840** may measure temperature of (or within) one or more of cores **808a**, **808b**, **808c**, voltage regulator **814**, memory **830**, a mother-board of SOC **801**, and/or any appropriate component of device **800**.

In some embodiments, device **800** comprises power measurement circuitries **842**, e.g., for measuring power consumed by one or more components of the device **800**. In an example, in addition to, or instead of, measuring power, the power measurement circuitries **842** may measure voltage and/or current. In an example, the power measurement circuitries **842** may be embedded, or coupled or attached to various components, whose power, voltage, and/or current consumption are to be measured and monitored. For example, power measurement circuitries **842** may measure power, current and/or voltage supplied by one or more voltage regulators **814**, power supplied to SOC **801**, power

supplied to device **800**, power consumed by processor **804** (or any other component) of device **800**, etc.

In some embodiments, device **800** comprises one or more voltage regulator circuitries, generally referred to as voltage regulator (VR) **814**. VR **814** generates signals at appropriate voltage levels, which may be supplied to operate any appropriate components of the device **800**. Merely as an example, VR **814** is illustrated to be supplying signals to processor **804** of device **800**. In some embodiments, VR **814** receives one or more Voltage Identification (VID) signals, and generates the voltage signal at an appropriate level, based on the VID signals. Various type of VRs may be utilized for the VR **814**. For example, VR **814** may include a “buck” VR, “boost” VR, a combination of buck and boost VRs, low dropout (LDO) regulators, switching DC-DC regulators, etc. Buck VR is generally used in power delivery applications in which an input voltage needs to be transformed to an output voltage in a ratio that is smaller than unity. Boost VR is generally used in power delivery applications in which an input voltage needs to be transformed to an output voltage in a ratio that is larger than unity. In some embodiments, each processor core has its own VR which is controlled by PCU **810a/b** and/or PMIC **812**. In some embodiments, each core has a network of distributed LDOs to provide efficient control for power management. The LDOs can be digital, analog, or a combination of digital or analog LDOs.

In some embodiments, device **800** comprises one or more clock generator circuitries, generally referred to as clock generator **816**. Clock generator **816** generates clock signals at appropriate frequency levels, which may be supplied to any appropriate components of device **800**. Merely as an example, clock generator **816** is illustrated to be supplying clock signals to processor **804** of device **800**. In some embodiments, clock generator **816** receives one or more Frequency Identification (FID) signals, and generates the clock signals at an appropriate frequency, based on the FID signals.

In some embodiments, device **800** comprises battery **818** supplying power to various components of device **800**. Merely as an example, battery **818** is illustrated to be supplying power to processor **804**. Although not illustrated in the figures, device **800** may comprise a charging circuitry, e.g., to recharge the battery, based on Alternating Current (AC) power supply received from an AC adapter.

In some embodiments, device **800** comprises Power Control Unit (PCU) **810** (also referred to as Power Management Unit (PMU), Power Controller, etc.). In an example, some sections of PCU **810** may be implemented by one or more processing cores **808**, and these sections of PCU **810** are symbolically illustrated using a dotted box and labelled PCU **810a**. In an example, some other sections of PCU **810** may be implemented outside the processing cores **808**, and these sections of PCU **810** are symbolically illustrated using a dotted box and labelled as PCU **810b**. PCU **810** may implement various power management operations for device **800**. PCU **810** may include hardware interfaces, hardware circuitries, connectors, registers, etc., as well as software components (e.g., drivers, protocol stacks), to implement various power management operations for device **800**.

In some embodiments, device **800** comprises Power Management Integrated Circuit (PMIC) **812**, e.g., to implement various power management operations for device **800**. In some embodiments, PMIC **812** is a Reconfigurable Power Management ICs (RPMICs) and/or an IMVP (Intel® Mobile Voltage Positioning). In an example, the PMIC is within an IC chip separate from processor **804**. PMIC **812** may implement various power management operations for device **800**.

PMIC **812** may include hardware interfaces, hardware circuitries, connectors, registers, etc., as well as software components (e.g., drivers, protocol stacks), to implement various power management operations for device **800**.

In an example, device **800** comprises one or both PCU **810** or PMIC **812**. In an example, any one of PCU **810** or PMIC **812** may be absent in device **800**, and hence, these components are illustrated using dotted lines.

Various power management operations of device **800** may be performed by PCU **810**, by PMIC **812**, or by a combination of PCU **810** and PMIC **812**. For example, PCU **810** and/or PMIC **812** may select a power state (e.g., P-state) for various components of device **800**. For example, PCU **810** and/or PMIC **812** may select a power state (e.g., in accordance with the ACPI (Advanced Configuration and Power Interface) specification) for various components of device **800**. Merely as an example, PCU **810** and/or PMIC **812** may cause various components of the device **800** to transition to a sleep state, to an active state, to an appropriate C state (e.g., C0 state, or another appropriate C state, in accordance with the ACPI specification), etc. In an example, PCU **810** and/or PMIC **812** may control a voltage output by VR **814** and/or a frequency of a clock signal output by the clock generator, e.g., by outputting the VID signal and/or the FID signal, respectively. In an example, PCU **810** and/or PMIC **812** may control battery power usage, charging of battery **818**, and features related to power saving operation.

The clock generator **816** can comprise a phase locked loop (PLL), frequency locked loop (FLL), or any suitable clock source. In some embodiments, each core of processor **804** has its own clock source. As such, each core can operate at a frequency independent of the frequency of operation of the other core. In some embodiments, PCU **810** and/or PMIC **812** performs adaptive or dynamic frequency scaling or adjustment. For example, clock frequency of a processor core can be increased if the core is not operating at its maximum power consumption threshold or limit. In some embodiments, PCU **810** and/or PMIC **812** determines the operating condition of each core of a processor, and opportunistically adjusts frequency and/or power supply voltage of that core without the core clocking source (e.g., PLL of that core) losing lock when the PCU **810** and/or PMIC **812** determines that the core is operating below a target performance level. For example, if a core is drawing current from a power supply rail less than a total current allocated for that core or processor **804**, then PCU **810** and/or PMIC **812** can temporarily increase the power draw for that core or processor **804** (e.g., by increasing clock frequency and/or power supply voltage level) so that the core or processor **804** can perform at a higher performance level. As such, voltage and/or frequency can be increased temporally for processor **804** without violating product reliability.

In an example, PCU **810** and/or PMIC **812** may perform power management operations, e.g., based at least in part on receiving measurements from power measurement circuitries **842**, temperature measurement circuitries **840**, charge level of battery **818**, and/or any other appropriate information that may be used for power management. To that end, PMIC **812** is communicatively coupled to one or more sensors to sense/detect various values/variations in one or more factors having an effect on power/thermal behavior of the system/platform. Examples of the one or more factors include electrical current, voltage droop, temperature, operating frequency, operating voltage, power consumption, inter-core communication activity, etc. One or more of these sensors may be provided in physical proximity (and/or thermal contact/coupling) with one or more components or

logic/IP blocks of a computing system. Additionally, sensor(s) may be directly coupled to PCU **810** and/or PMIC **812** in at least one embodiment to allow PCU **810** and/or PMIC **812** to manage processor core energy at least in part based on value(s) detected by one or more of the sensors.

Also illustrated is an example software stack of device **800** (although not all elements of the software stack are illustrated). Merely as an example, processors **804** may execute application programs **850**, Operating System **852**, one or more Power Management (PM) specific application programs (e.g., generically referred to as PM applications **858**), and/or the like. PM applications **858** may also be executed by the PCU **810** and/or PMIC **812**. OS **852** may also include one or more PM applications **856a**, **856b**, **856c**. The OS **852** may also include various drivers **854a**, **854b**, **854c**, etc., some of which may be specific for power management purposes. In some embodiments, device **800** may further comprise a Basic Input/Output System (BIOS) **820**. BIOS **820** may communicate with OS **852** (e.g., via one or more drivers **854**), communicate with processors **804**, etc.

For example, one or more of PM applications **858**, **856**, drivers **854**, BIOS **820**, etc. may be used to implement power management specific tasks, e.g., to control voltage and/or frequency of various components of device **800**, to control wake-up state, sleep state, and/or any other appropriate power state of various components of device **800**, control battery power usage, charging of the battery **818**, features related to power saving operation, etc.

In some embodiments, device **800** comprises module circuit board **860** and bus **862**. Module circuit board **860** may be an add-in card or an expansion card comprising a connector **118**. Module circuit board **860** may comprise a module and module circuitry (not shown). In various embodiments, module circuit board **860** may be similar to or the same as module circuit board **108**, **208**, or **508**. In various embodiments, the module may be similar to or the same as module **104**, **204**, or **504**, respectively. In embodiments, bus **862** may be a PCIe bus, and may be the same as or similar to bus **106**, **206**, or **506**. Bus **862** may conform to an industry specification, e.g. PCIe. In embodiments, bus **862** comprises a socket to receive a connector of module circuit board **860**.

In some embodiments, module circuit board **860** and control hub **832** communicate to limit electric current consumption that module circuit board **860** receives power from bus **862**. In these embodiments, module circuit board **860** comprises instances of first sampler circuitry **128**, utilization monitoring and power estimating circuitry **130**, and capability setting circuitry **132** (not shown). In addition, control hub **832** comprises an instance of first sampler circuitry **128** (not shown).

In some embodiments, module circuit board **860** and control hub **832** communicate to assure that the voltage level used by module circuit board **860** and by control hub **832** for sideband signaling are compatible, e.g., match. In these embodiments, module circuit board **860** comprises an instance of second notifier **536**, control hub **832** comprises an instance of second sampler circuitry **517**, and device **800** comprises one or more instances of voltage level translator **538**, **539**.

Furthermore, the particular features, structures, functions, or characteristics may be combined in any suitable manner in one or more embodiments. For example, a first embodiment may be combined with a second embodiment anywhere the particular features, structures, functions, or characteristics associated with the two embodiments are not mutually exclusive.

While the disclosure has been described in conjunction with specific embodiments thereof, many alternatives, modifications and variations of such embodiments will be apparent to those of ordinary skill in the art in light of the foregoing description. The embodiments of the disclosure are intended to embrace all such alternatives, modifications, and variations as to fall within the broad scope of the appended claims.

In addition, well-known power/ground connections to integrated circuit (IC) chips and other components may or may not be shown within the presented figures, for simplicity of illustration and discussion, and so as not to obscure the disclosure. Further, arrangements may be shown in block diagram form in order to avoid obscuring the disclosure, and also in view of the fact that specifics with respect to implementation of such block diagram arrangements are highly dependent upon the platform within which the present disclosure is to be implemented (i.e., such specifics should be well within purview of one skilled in the art). Where specific details (e.g., circuits) are set forth in order to describe example embodiments of the disclosure, it should be apparent to one skilled in the art that the disclosure can be practiced without, or with variation of, these specific details. The description is thus to be regarded as illustrative instead of limiting.

The following examples pertain to further embodiments. Specifics in the examples may be used anywhere in one or more embodiments. All optional features of the apparatus described herein may also be implemented with respect to a method or process. The examples can be combined in any combinations. For example, example 4 can be combined with example 2.

Example 1: An apparatus comprising: a circuit board comprising a connector in or on the circuit board, wherein the apparatus is to be coupled to a bus via the connector; first circuitry coupled to the circuit board, the first circuitry to participate in a communication with a processor via the connector and the bus; second circuitry coupled to the first circuitry, the second circuitry to: detect a utilization state of the first circuitry; determine, based on the detected utilization state of the first circuitry, a level of current to be conducted with the connector; and generate a signal which comprises an indication of whether the level of current exceeds a threshold current capacity of the connector; and third circuitry coupled to the first circuitry and the second circuitry, wherein, based on the signal, the third circuitry is to select a first operational mode from among multiple operational modes, wherein the first circuitry is to operate in any of the multiple operational modes responsive to the third circuitry.

Example 2: The apparatus of example 1, further comprising: fourth circuitry coupled to the circuit board, the fourth circuitry to participate in another communication via the connector and the bus, the other communication to identify a capability of the apparatus to accommodate a voltage level at the connector.

Example 3: The apparatus of example 2, further comprising: fifth circuitry coupled to the fourth circuitry to determine whether the capability of the apparatus to accommodate a voltage level at the connector is compatible with capability of the processor to accommodate a voltage level at the connector; and sixth circuitry, in response to determining that the capability of the apparatus to accommodate a voltage level at the connector is not compatible with capability of the processor to accommodate a voltage level at the connector, to configure voltage level shifting circuitry connected to the bus.

Example 4: The apparatus of example 2, wherein the connector comprises a plurality of interconnect structures, each interconnect structure corresponding with a bus line of the bus, and the other communication is to identify a capability of the apparatus to accommodate a voltage level at one or more of the interconnect structures of the connector, and wherein the one or more of the interconnect structures of the connector are to be used for a sideband signal comprising a full card power off signal, a module reset signal, a wireless disable signal, a co-existence signal, or a configuration signal.

Example 5: The apparatus of example 1, further comprising: fourth circuitry coupled to the circuit board, the fourth circuitry to participate in another communication via the connector and the bus, the other communication to identify the threshold current capacity of the connector.

Example 6: The apparatus of example 1, wherein operation of the first circuitry in the first operational mode is associated with a level of current in the conductor that is less than or equal to the threshold current capacity of the connector.

Example 7: The apparatus of example 6, wherein the first circuitry is a wireless wide area network (WWAN) module and the first operational mode uses a particular number of component carrier connections or a particular transmit power value.

Example 8: The apparatus of example 6, wherein the first circuitry is a solid state drive (SSD) memory module and the first operational mode uses a particular dynamic power allocation (DPA) sub-state.

Example 9: The apparatus of example 6, wherein the first circuitry is a graphics processing module, and the first operational mode uses a particular clock value.

Example 10: The apparatus of example 1, wherein the connector is a peripheral component interconnect express (PCIe) M.2 form factor connector.

Example 11: The apparatus of example 1, wherein the bus is a peripheral component interconnect express (PCIe) bus.

Example 12: A system comprising: a processor; a module coupled to the processor via a bus, the module comprising: a circuit board, wherein the module is coupled to the bus via a connector in or on a circuit board; first circuitry coupled to the circuit board, the first circuitry to participate in a communication with the processor via the connector and the bus; second circuitry coupled to the first circuitry, the second circuitry to: determine, based on a utilization state of the first circuitry, a level of current to be conducted with the connector, and generate a signal which comprises an indication of whether the level of current exceeds a threshold current capacity of the connector; and third circuitry coupled to the first circuitry and the second circuitry, wherein, based on the signal, the third circuitry is to select a first operational mode from among multiple operational modes, wherein the first circuitry is to operate in any of the multiple operational modes responsive to the third circuitry.

Example 13: The system of example 12, further comprising: fourth circuitry coupled to the circuit board, the fourth circuitry to participate in another communication via the connector and the bus, the other communication to identify a capability of the module to accommodate a voltage level at the connector; fifth circuitry coupled to the fourth circuitry to determine whether the capability of the module to accommodate a voltage level at the connector is compatible with capability of the processor to accommodate a voltage level at the connector; and sixth circuitry, in response to determining that the capability of the module to accommodate a voltage level at the connector is not compatible with

capability of the processor to accommodate a voltage level at the connector, to configure voltage level shifting circuitry connected to the bus.

Example 14: The system of example 12, further comprising: fourth circuitry coupled to the circuit board, the fourth circuitry to participate in another communication via the connector and the bus, the other communication to identify the threshold current capacity of the connector.

Example 15: The system of example 12, wherein operation of the first circuitry in the first operational mode is associated with a level of current in the conductor that is less than or equal to the threshold current capacity of the connector.

Example 16: The system of example 12, wherein the module comprises a wide-area network (WWAN) module, a graphics module, or a solid-state memory device (SSD).

Example 17: A packaged device comprising: a hardware interface to couple the packaged device to a circuit board; first circuitry coupled to the hardware interface, the first circuitry to participate in a communication with a processor while the circuit board is coupled to the processor via a bus and a connector in or on the circuit board; second circuitry coupled to the first circuitry, the second circuitry to: determine a threshold current capacity of the connector; detect a utilization state of the first circuitry; determine, based on the detected utilization state of the first circuitry, a level of current to be conducted with the connector; and generate a signal which comprises an indication of whether the level of current exceeds the threshold current capacity of the connector; and third circuitry coupled to the first circuitry and the second circuitry, wherein, based on the signal, the third circuitry is to select a first operational mode from among multiple operational modes, wherein the first circuitry is to operate in any of the multiple operational modes responsive to the third circuitry.

Example 18: The packaged device of example 17, further comprising: fourth circuitry coupled to the circuit board, the fourth circuitry to participate in another communication via the connector and the bus, the other communication to identify a capability of the packaged device to accommodate a voltage level at the connector.

Example 19: The packaged device of example 17, further comprising: fifth circuitry coupled to the fourth circuitry to determine whether the capability of the packaged device to accommodate a voltage level at the connector is compatible with capability of the processor to accommodate a voltage level at the connector; and sixth circuitry, in response to determining that the capability of the packaged device to accommodate a voltage level at the connector is not compatible with capability of the processor to accommodate a voltage level at the connector, to configure voltage level shifting circuitry connected to the bus.

Example 20: The packaged device of example 17, further comprising: fourth circuitry coupled to the circuit board, the fourth circuitry to participate in another communication via the connector and the bus, the other communication to identify the threshold current capacity of the connector, wherein operation of the first circuitry in the first operational mode is associated with a level of current in the conductor that is less than or equal to the threshold current capacity of the connector.

An abstract is provided that will allow the reader to ascertain the nature and gist of the technical disclosure. The abstract is submitted with the understanding that it will not be used to limit the scope or meaning of the claims. The

following claims are hereby incorporated into the detailed description, with each claim standing on its own as a separate embodiment.

What is claimed is:

1. An apparatus comprising:

a circuit board comprising a connector in or on the circuit board, wherein the apparatus is to be coupled to a bus via the connector;

first circuitry coupled to the circuit board, the first circuitry to participate in a communication with a processor via the connector and the bus;

second circuitry coupled to the first circuitry, the second circuitry to:

detect a utilization state of the first circuitry;

determine, based on the detected utilization state of the first circuitry, a level of current to be conducted with the connector; and

generate a signal which comprises an indication of whether the level of current exceeds a threshold current capacity of the connector; and

third circuitry coupled to the first circuitry and the second circuitry, wherein, based on the signal, the third circuitry is to select a first operational mode from among multiple operational modes, wherein the first circuitry is to operate in any of the multiple operational modes responsive to the third circuitry.

2. The apparatus of claim 1, further comprising:

fourth circuitry coupled to the circuit board, the fourth circuitry to participate in another communication via the connector and the bus, the other communication to identify a capability of the apparatus to accommodate a voltage level at the connector.

3. The apparatus of claim 2, further comprising:

fifth circuitry coupled to the fourth circuitry to determine whether the capability of the apparatus to accommodate a voltage level at the connector is compatible with a capability of the processor to accommodate a voltage level at the connector; and

sixth circuitry, in response to determining that the capability of the apparatus to accommodate a voltage level at the connector is not compatible with the capability of the processor to accommodate a voltage level at the connector, to configure voltage level shifting circuitry connected to the bus.

4. The apparatus of claim 2, wherein the connector comprises a plurality of interconnect structures, each interconnect structure corresponding with a bus line of the bus, and the other communication is to identify a capability of the apparatus to accommodate a voltage level at one or more of the interconnect structures of the connector, and

wherein the one or more of the interconnect structures of the connector are to be used for a sideband signal comprising a full card power off signal, a module reset signal, a wireless disable signal, a co-existence signal, or a configuration signal.

5. The apparatus of claim 1, further comprising:

fourth circuitry coupled to the circuit board, the fourth circuitry to participate in another communication via the connector and the bus, the other communication to identify the threshold current capacity of the connector.

6. The apparatus of claim 1, wherein operation of the first circuitry in the first operational mode is associated with a level of current in the conductor that is less than or equal to the threshold current capacity of the connector.

7. The apparatus of claim 6, wherein the first circuitry is a wireless wide area network (WWAN) module and the first

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operational mode uses a particular number of component carrier connections or a particular transmit power value.

8. The apparatus of claim 6, wherein the first circuitry is a solid state drive (SSD) memory module and the first operational mode uses a particular dynamic power allocation (DPA) sub-state.

9. The apparatus of claim 6, wherein the first circuitry is a graphics processing module, and the first operational mode uses a particular clock value.

10. The apparatus of claim 1, wherein the connector is a peripheral component interconnect express (PCIe) M.2 form factor connector.

11. The apparatus of claim 1, wherein the bus is a peripheral component interconnect express (PCIe) bus.

12. A system comprising:

a processor;

a module coupled to the processor via a bus, the module comprising:

a circuit board, wherein the module is coupled to the bus via a connector in or on a circuit board;

first circuitry coupled to the circuit board, the first circuitry to participate in a communication with the processor via the connector and the bus;

second circuitry coupled to the first circuitry, the second circuitry to: determine, based on a utilization state of the first circuitry, a level of current to be conducted with the connector, and generate a signal which comprises an indication of whether the level of current exceeds a threshold current capacity of the connector; and

third circuitry coupled to the first circuitry and the second circuitry, wherein, based on the signal, the third circuitry is to select a first operational mode from among multiple operational modes, wherein the first circuitry is to operate in any of the multiple operational modes responsive to the third circuitry.

13. The system of claim 12, further comprising:

fourth circuitry coupled to the circuit board, the fourth circuitry to participate in another communication via the connector and the bus, the other communication to identify a capability of the module to accommodate a voltage level at the connector;

fifth circuitry coupled to the fourth circuitry to determine whether the capability of the module to accommodate a voltage level at the connector is compatible with a capability of the processor to accommodate a voltage level at the connector; and

sixth circuitry, in response to determining that the capability of the module to accommodate a voltage level at the connector is not compatible with the capability of the processor to accommodate a voltage level at the connector, to configure voltage level shifting circuitry connected to the bus.

14. The system of claim 12, further comprising:

fourth circuitry coupled to the circuit board, the fourth circuitry to participate in another communication via the connector and the bus, the other communication to identify the threshold current capacity of the connector.

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15. The system of claim 12, wherein operation of the first circuitry in the first operational mode is associated with a level of current in the conductor that is less than or equal to the threshold current capacity of the connector.

16. The system of claim 12, wherein the module comprises a wide-area network (WWAN) module, a graphics module, or a solid-state memory device (SSD).

17. A packaged device comprising:

first circuitry coupled to a hardware interface, the hardware interface to couple the packaged device to a circuit board, the first circuitry to participate in a communication with a processor while the circuit board is coupled to the processor via a bus and a connector; second circuitry coupled to the first circuitry, the second circuitry to:

determine a threshold current capacity of the connector; detect a utilization state of the first circuitry;

determine, based on the detected utilization state of the first circuitry, a level of current to be conducted with the connector; and

generate a signal which comprises an indication of whether the level of current to be conducted exceeds the threshold current capacity of the connector; and

third circuitry coupled to the first circuitry and the second circuitry, wherein, based on the signal, the third circuitry is to select a first operational mode from among multiple operational modes, wherein the first circuitry is to operate in any of the multiple operational modes responsive to the third circuitry.

18. The packaged device of claim 17, further comprising: fourth circuitry coupled to the circuit board, the fourth circuitry to participate in another communication via the connector and the bus, the other communication to identify a capability of the packaged device to accommodate a voltage level at the connector.

19. The packaged device of claim 18, further comprising: fifth circuitry coupled to the fourth circuitry to determine whether the capability of the packaged device to accommodate a voltage level at the connector is compatible with capability of the processor to accommodate a voltage level at the connector; and

sixth circuitry, in response to determining that the capability of the packaged device to accommodate a voltage level at the connector is not compatible with a capability of the processor to accommodate a voltage level at the connector, to configure voltage level shifting circuitry connected to the bus.

20. The packaged device of claim 17, further comprising: fourth circuitry coupled to the circuit board, the fourth circuitry to participate in another communication via the connector and the bus, the other communication to identify the threshold current capacity of the connector, wherein operation of the first circuitry in the first operational mode is associated with a level of current in the conductor that is less than or equal to the threshold current capacity of the connector.

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